The Coversheet

Student Name (unless anonymised)	
Student Number (as shown on student ID card):	**240174581**
Word Count / Pages / Duration / Other Limits:	4707
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Date of Submission:	**27th May 2025, 12Noon**

I have read and understood the Academic Misconduct statement.	Tick to confirm ☒
I have read and understood the Generative Artificial Intelligence use statement.	Tick to confirm ☒
I am satisfied that I have met the Learning Outcomes of this assignment (please check the Assignment Brief if you are unsure)	Met ☒

Self-Assessment – If there are aspects of your assignment on which you would like feedback, please indicate below.

Optional for students

Suggested prompt questions-

How have you developed or progressed you're learning in this work?
What do you feel is the strongest part of this submission?
What feedback would you give yourself?
What part(s) of this assignment are you still unsure about?

- Using Python, I have improved my capacity to clean and examine real-world healthcare data as well as apply several visualization techniques to derive practical insights. Managing sensitive data helped me also better grasp ethical and legal obligations.
- The use of sophisticated visualizations like the sunburst and animated bar charts—which offer clear, multi-layered insights into seasonal trends, gender-specific health issues, and healthcare costs—is the strongest aspect.
- I would give myself credit for successfully fusing technical analysis with lucid visual narrative. My work demonstrates a strong command of healthcare data interpretation as well as Python-based visualization tools. To improve the overall impact of my work, I would challenge myself moving forward to investigate more diverse datasets and push the boundaries of interactive dashboards or predictive analytics.
- Particularly with relation to pseudonymisation and data sharing rights, I still find it rather difficult to understand and apply legal frameworks like UK GDPR in complicated situations.

Assessor's Feedback (may be delivered in line with the submission)

Were the learning outcomes met?	Yes ☒ If not, what was not met:
Assessor's response to the student's submission, request for feedback and / or self-assessment (feedback):	
What specific actions should the student undertake to progress their learning? (feedforward):	
Please take this and other feedback to your next academic tutorial to plan your future work.	

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CHAPTER ONE

INTRODUCTION

This assessment intends to investigate important healthcare discoveries by means of data visualization methods over a Kaggle Healthcare Dataset. By converting difficult data into clear, useful illustrations, data visualization aids in comprehending vast datasets (Few, 2007). Comprising 55,500 rows and 15 attributes, the collection covers patient demographics, medical conditions, hospital information, and billing information. It was selected since, in the critical sector of healthcare, data-driven insights can enhance planning for finances, hospital administration, and care for patients. The task at hand aims to produce useful graphs that underline patterns in medical costs, admissions to hospitals, and disease distribution. This guarantees informed judgments can be made by researchers, legislators, and medical practitioners. Visualizing patterns and connections in the data helps us to spot important medical trends and abnormalities (Hudiburgh & Garbinsky, 2020).

The paper will show how data visualization improves healthcare analytics by including data cleaning, visualizing techniques, and important conclusions.

CHAPTER TWO

DATASET OVERVIEW AND SELECTION

Source of Dataset: The dataset utilized in this assessment was taken from Kaggle, a popular platform for providing datasets and constructing machine learning algorithms. The dataset is titled “Healthcare_Dataset” and is managed by Prasad Patil. It can be found at: https://www.kaggle.com/datasets/prasad22/healthcare-dataset (Patil, 2021).

Dataset Domain: This dataset relates to the health care domain, focused on hospital patient information. It comprises both clinical and administrative features such as patient demographics, medical problems, admission records, billing, and insurance details. These criteria are critical for health care data analytics and hospital efficiency tracking.

Dataset Description: The dataset has 55,501 rows and 15 characteristics, where each row stands for a distinct patient admission history. The characteristics include demographic details such as “Name”, “Age”, “Gender”, and “Blood Type”, healthcare data like “Medical Condition”, “Date of Admission”, and “Doctor”, and “administrative and financial details” including “Hospital”, “Insurance Provider”, “Billing Amount”, “Room Number”, and “Admission Type”. These features give a broad picture of the patient experience, from medical diagnosis to management processing, making the dataset ideal for both statistical and graphical study. The mix of qualitative and numerical data enables varied data visualization methods, enabling the detection of crucial overall trends in healthcare.

Selection Justification: The dataset was selected for its depth, amount, and several attributes, representing real-world hospital information. Its mix of clinical and non-clinical characteristics enables useful study of patient demographic information, disease patterns, healthcare expenditures, and operational indicators. With over 55,000 rows, it provides solid graphical narrative and insight production. In addition, healthcare is a socially significant subject, where data-driven findings can benefit service delivery, patient results, and resource distribution. This coincides with the primary purpose of data visualization—to convey complicated information clearly, precisely, and effectively.

CHAPTER THREE
DATA CLEANING AND PREPROCESSING

Successful data graphical representation begins with extensive data cleaning and preparation, as unprepared datasets often contain missing values, errors, and oddities that can mislead studies. For this study, many measures were conducted to ensure the dataset was appropriate for visualization and analysis.

Initial Data Inspection: Data cleaning, or data cleansing, is the process of discovering and resolving flaws, inconsistencies, and inaccuracies in datasets. It assures the data you deal with is accurate, dependable, and ready for analysis, visualization, or additional processing. It is a vital stage in the data preparation pipeline (LearnPython.com, 2023).

The dataset was loaded into python jupyter notebook environment using "**df = pd.read_csv("healthcare_dataset.csv")"**

```
df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 55500 entries, 0 to 55499
Data columns (total 15 columns):
 #   Column              Non-Null Count  Dtype  
---  ------              --------------  -----  
 0   Name                55500 non-null  object 
 1   Age                 55500 non-null  int64  
 2   Gender              55500 non-null  object 
 3   Blood Type          55500 non-null  object 
 4   Medical Condition   55500 non-null  object 
 5   Date of Admission   55500 non-null  object 
 6   Doctor              55500 non-null  object 
 7   Hospital            55500 non-null  object 
 8   Insurance Provider  55500 non-null  object 
 9   Billing Amount      55500 non-null  float64
 10  Room Number         55500 non-null  int64  
 11  Admission Type      55500 non-null  object 
 12  Discharge Date      55500 non-null  object 
 13  Medication          55500 non-null  object 
 14  Test Results        55500 non-null  object 
dtypes: float64(1), int64(2), object(12)
memory usage: 6.4+ MB
df.shape
(55500, 15)
```

Figure 1: Dataset Information

From the figure above, we have **rows=55500**, **columns =15,** there is no missing values recorded, and the datatypes are **"int64"**, **"float64"** and **"object"**

Standardizing Columns: As part of the data cleaning process, the following text features were standardized to proper format which are **"Name, Gender, Doctor, Hospital, Insurance Provider, Medical Condition, Admission Type, Medication, Test Results, Blood Type".**

The date features were converted to Datetime format and duplicate values were dropped and the shape before and after were given as: **Before = (55500, 15). After = (54860, 15)**.

Outlier Check Using IQR Method: for the **"age"**: there are no excessive or unreasonable age values since all patient ages fit a reasonable range **(0 to 117 years)**. For the **"Billing Amount"**: your collection has no negative billing values even if the lower bound is negative. This implies: no credit abnormalities or unanticipated returns. Billing levels are within regular hospital guidelines. Financial data makes sense. For the **"Room Number"**: Once more, the computed lower bound is negative; but your data does not show any real negative room counts. This suggests that every room number—including Room 1–699 is within a reasonable range. Room assignments follow natural logic.

```
df.head()
```

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
0	Bobby Jackson	30	Male	B-	Cancer	2024-01-31	Matthew Smith	Sons And Miller	Blue Cross	18856.28	328	Urgent	2024-02-02	Paracetamol	Normal
1	Leslie Terry	62	Male	A+	Obesity	2019-08-20	Samantha Davies	Kim Inc	Medicare	33643.33	265	Emergency	2019-08-26	Ibuprofen	Inconclusive
2	Danny Smith	76	Female	A-	Obesity	2022-09-22	Tiffany Mitchell	Cook Plc	Aetna	27955.10	205	Emergency	2022-10-07	Aspirin	Normal
3	Andrew Watts	28	Female	O+	Diabetes	2020-11-18	Kevin Wells	Hernandez Rogers And Vang,	Medicare	37909.78	450	Elective	2020-12-18	Ibuprofen	Abnormal
4	Adrienne Bell	43	Female	AB+	Cancer	2022-09-19	Kathleen Hanna	White-White	Aetna	14238.32	458	Urgent	2022-10-09	Penicillin	Abnormal

Figure 2: First few rows of the cleaned dataset

```
df.dtypes
```

```
Name                          object
Age                            int64
Gender                        object
Blood Type                    object
Medical Condition             object
Date of Admission     datetime64[ns]
Doctor                        object
Hospital                      object
Insurance Provider            object
Billing Amount               float64
Room Number                    int64
Admission Type                object
Discharge Date        datetime64[ns]
Medication                    object
Test Results                  object
dtype: object
```

Figure 3: Datatypes of the cleaned dataset features

CHAPTER FOUR
EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) acts as a key first stage in any data-driven investigation process. It entails summarizing the essential properties of the dataset, frequently using visual approaches, to comprehend the structure, find anomalies, test assumptions, and generate hypotheses. In the setting of healthcare data, EDA is especially crucial as it helps detect trends, correlations, and outliers that may affect patient results or impact diagnostic and billing choices (Katarya and Jain, 2020). Analysts study datasets to understand distribution

patterns, correlations, and potential data quality issues, enabling preprocessing, feature engineering, and model selection for improved analysis and machine learning performance (Zhao, 2022).

Exploratory Data Analysis (EDA) is a crucial process for initial investigations, identifying patterns, and checking assumptions using summary statistics and graphical representations. It helps identify meaningful patterns, provides in-depth insights, and imputes missing values (Mahadevan, 2022).

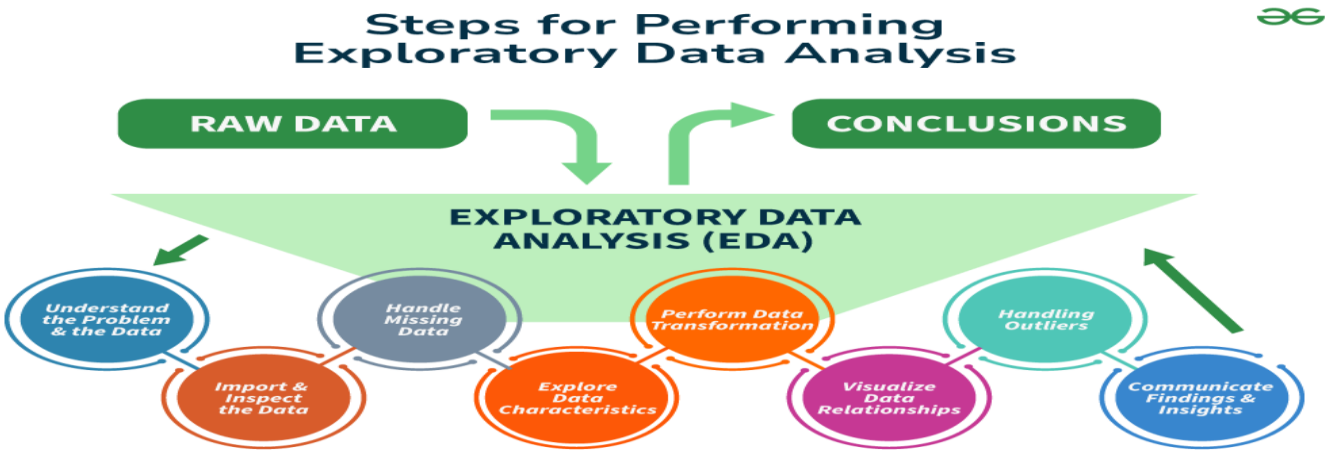


Figure 4: Steps for EDA (GeeksforGeeks, 2021).

Descriptive Statistics of the Dataset:

```
df.describe().T
```

	count	mean	min	25%	50%	75%	max	std
Age	54860.0	51.53385	13.0	35.0	52.0	68.0	89.0	19.605295
Date of Admission	54860	2021-11-01 19:01:50.506744576	2019-05-08 00:00:00	2020-07-28 00:00:00	2021-11-02 00:00:00	2023-02-03 00:00:00	2024-05-07 00:00:00	NaN
Billing Amount	54860.0	25594.633612	9.24	13299.75	25593.875	37847.065	52764.28	14175.867046
Room Number	54860.0	301.109752	101.0	202.0	302.0	400.0	500.0	115.217195
Discharge Date	54860	2021-11-17 07:00:08.137076224	2019-05-09 00:00:00	2020-08-13 00:00:00	2021-11-18 00:00:00	2023-02-19 00:00:00	2024-06-06 00:00:00	NaN

Figure 5: Descriptive Statistics of Numerical Columns.

From figure 5 above. The dataset is clear, balanced, and complete according to the descriptive statistics. Its adaptability for further exploratory study and machine learning purposes in healthcare is highlighted by the frequency of chronic diseases, range of admission kinds, and great fluctuation in billing amounts.

Univariate Analysis of Numerical columns:

The following figures below show the distribution of each of the numerical columns in the dataset, and they are:

- **Age:** mainly between 20 and 70 years, the distribution is rather right skewed, and this indicates a major concentration in adult to early year intervals.
- **Billing Amount:** this has a positive skewed distribution and with a few high-cost outliers over 50,000, most patients are charged between 10,000 and 30,000.

- **Room Number:** reasonably consistent spread between Room 100 and Room 500. indicates no bias in room distribution throughout departments.

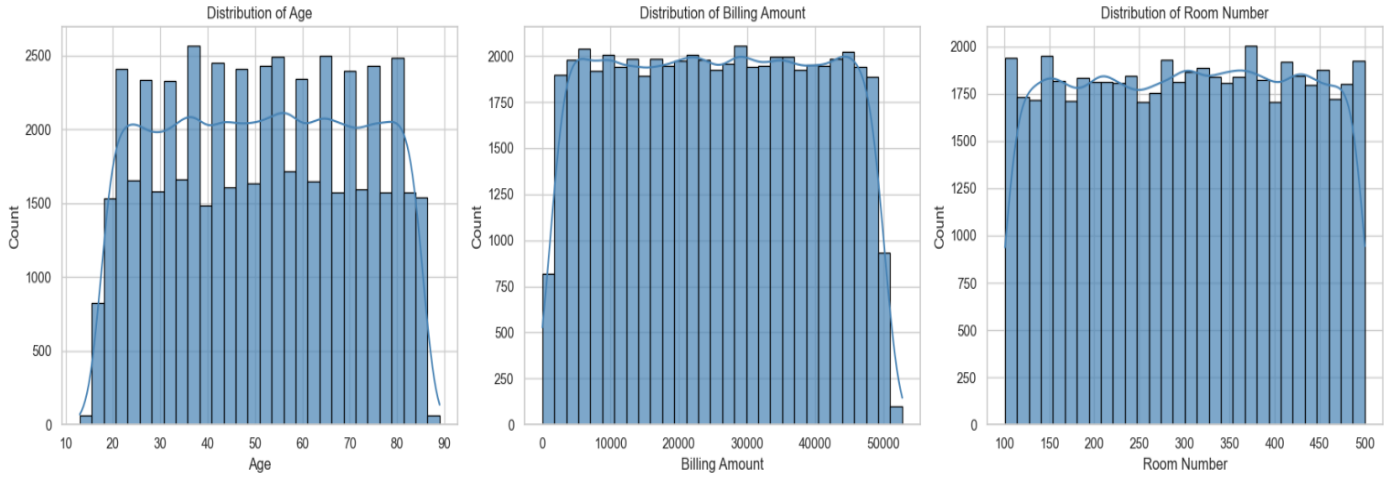
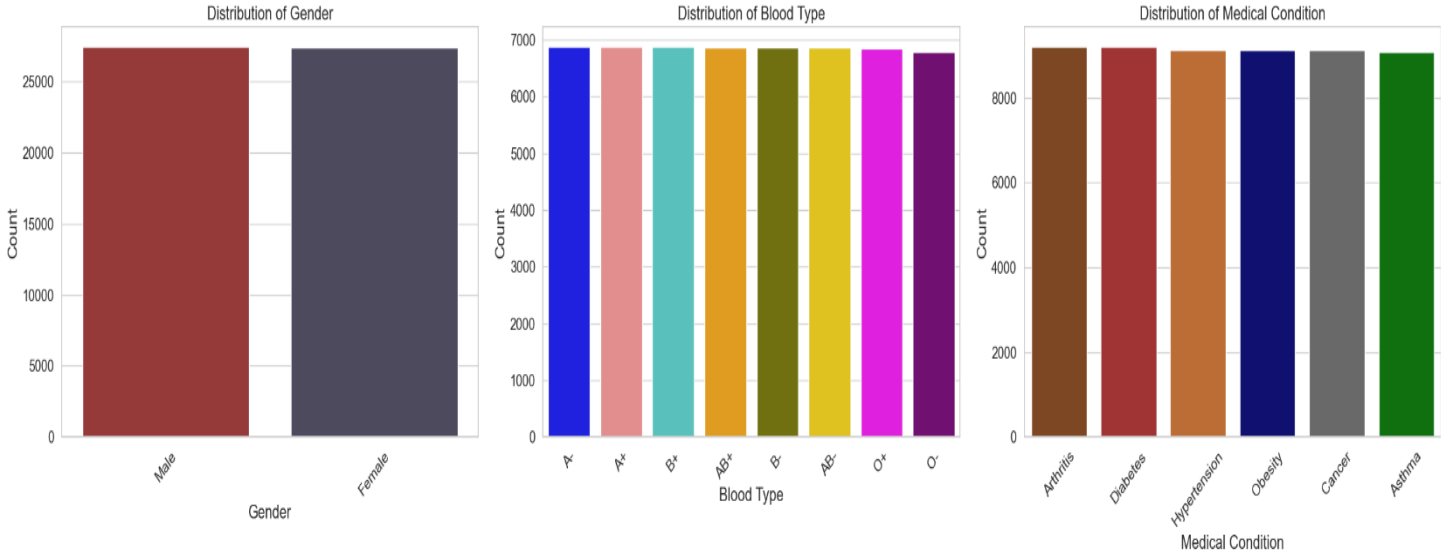


Figure 6: Distribution visualization of the Numerical features.

Univariate Analysis of Categorical columns:

The "**Name**" and **"Doctor"** variables were excluded in our univariate analysis because they have many unique data points.



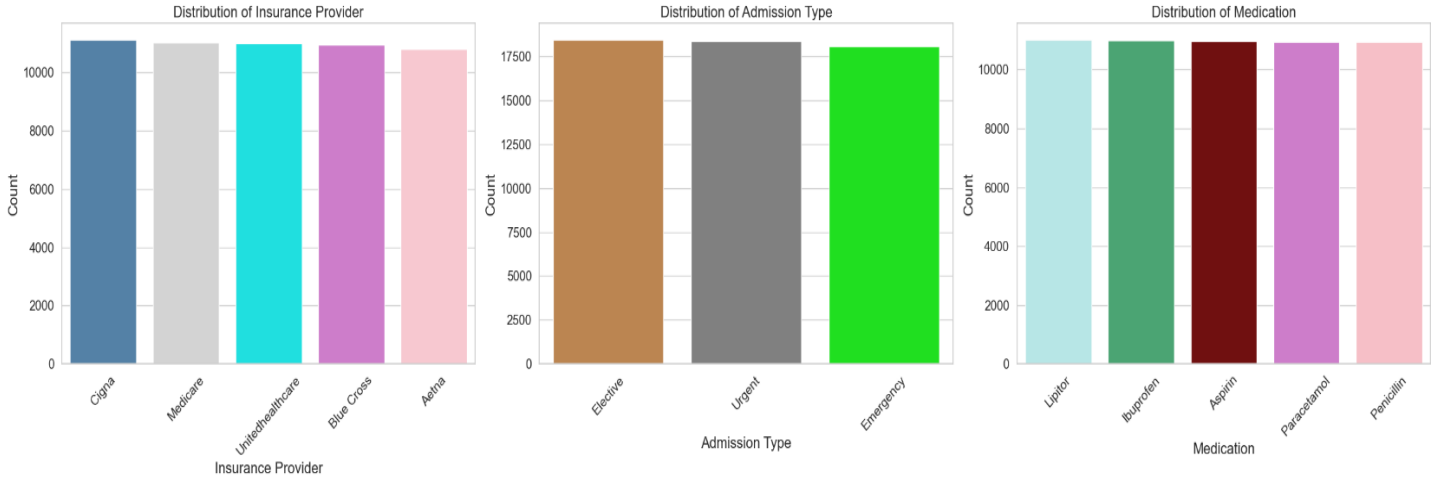




Figure 7: Bar graphs of the Categorical variables in the dataset.

The following insights were gathered from the bar graphs of the categorical features above:

- The hospital's service population exhibits a fair gender balance, with female patients being more common than male ones, ensuring generalizability in gender-based research.
- The dataset includes representations of major blood types, with O+ and A+ being the most frequently occurring, potentially impacting blood transfusion availability policies and treatment planning worldwide.
- Chronic lifestyle diseases, such as hypertension, diabetes, obesity, and asthma, are prevalent, emphasizing the need for ongoing population surveillance and preventive measures.
- Common drugs like Paracetamol, Ibuprofen, and Aspirin are used for pain relief and inflammation, while lower numbers for Amoxicillin and Penicillin may indicate fewer bacterial infections.
- Most patients are admitted urgently, indicating a higher demand for acute care, suggesting the need for public health awareness, early screening, and preventative outreach.
- Most test outcomes are classified as "Normal," followed by "Abnormal" and "Inconclusive," suggesting a patient population requiring additional diagnostic attention and care.
- Patients primarily use Aetna and Blue Cross as major insurance suppliers, which could enhance hospital partnership prioritization, claims processing efficiency, and insurance service personalization.

Bivariate Analysis:

We shall view the relationship between **“Age”** and **“Billing Amount”** with the use of scattered plot.
The observations from the scatter plot in figure 8 below imply that:

- The data points are reasonably evenly distributed, showing no clear linear trend.
- There’s a tiny positive link, showing that billing tends to increase with age, though it’s not extremely obvious.
- Billing amount is very variable through all age groups, which shows that other factors e.g., medical condition or admission type may influence expenses

```
# Scatter plot that shows the relationship between Age and Billing amount
sns.set(style="whitegrid")
plt.figure(figsize=(10, 6))
sns.regplot(data=df, x='Age', y='Billing Amount', scatter_kws={'alpha':0.5}, line_kws={'color': 'red'})
plt.title('Relationship Between Age and Billing Amount', fontsize=14)
plt.xlabel('Age')
plt.ylabel('Billing Amount')
plt.tight_layout()
plt.show() # Display the plots
```

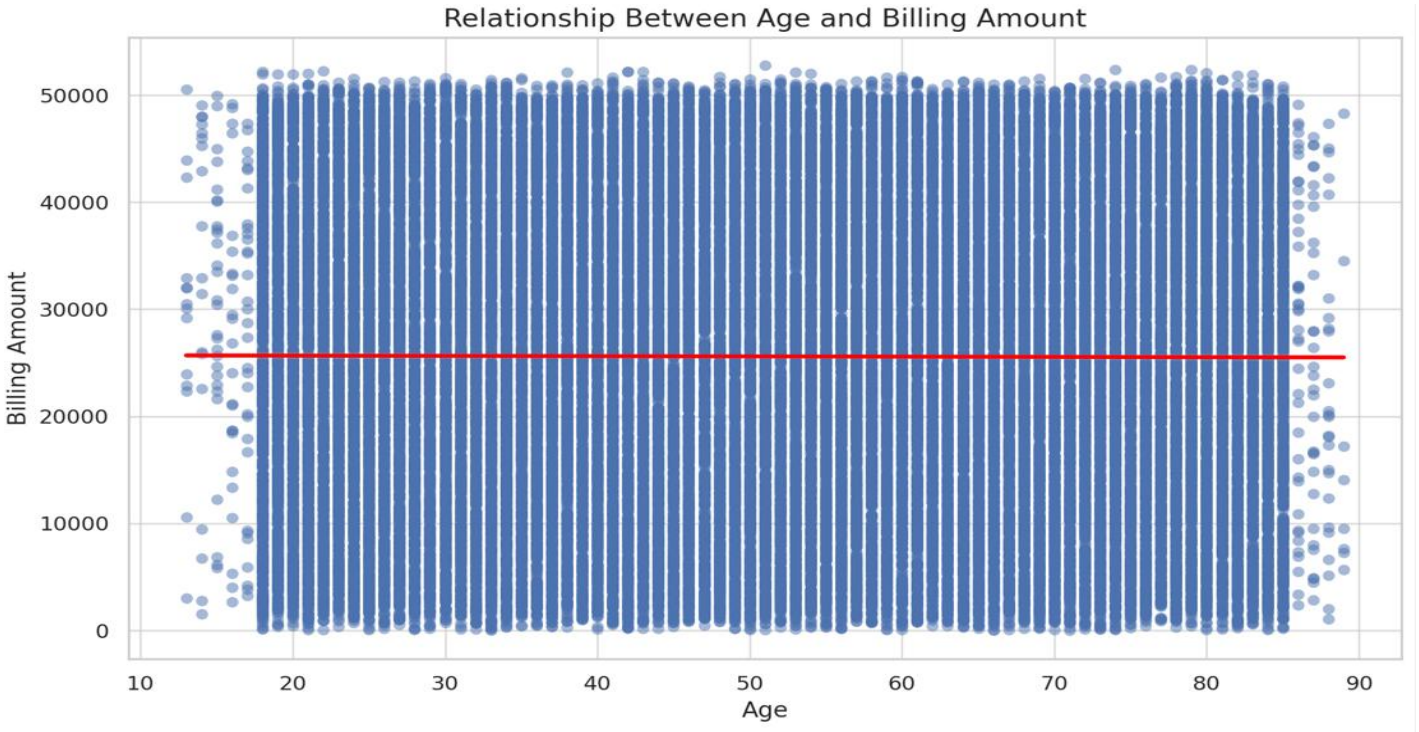


Figure 8: Relationship between Age and Billing Amount.

Violin Plot: This plot is used to show the relationship between Billing Amount and Medical Condition

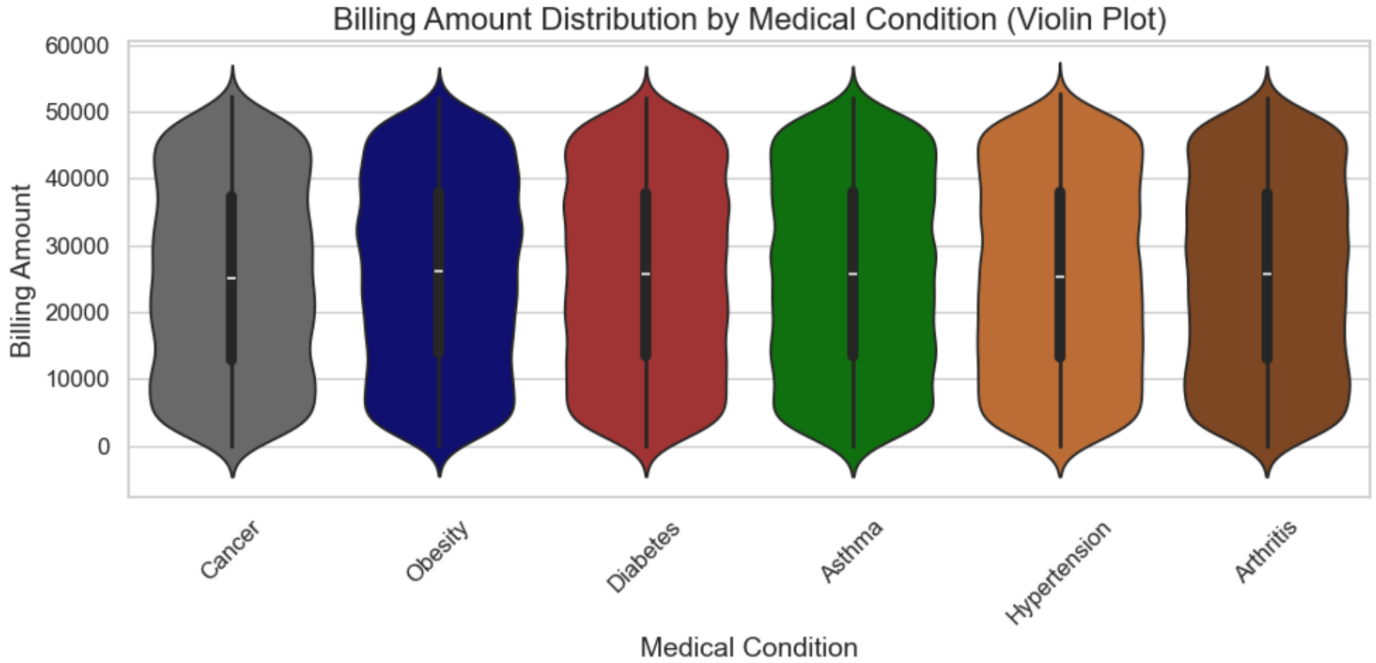


Figure 9: Billing Amount by Medical Condition

Key Insights from the Figure 9 above are:

- **Shape of Distribution:** The violin plots display a uniform distribution with no noticeable outliers, indicating a steady dip in values across all categories.
- **Range of Billing:** The billing values for all diseases range from $0 to $60,000, indicating a consistent treatment cost structure across different conditions, with median lines showing nearly correlated billing values.

- **Distribution width:** The larger the middle plot, the more patients fall within the billing range, with diseases like Asthma, Hypertension, and Diabetes showing wider centres.

Box plot of Billing Amount by Gender:

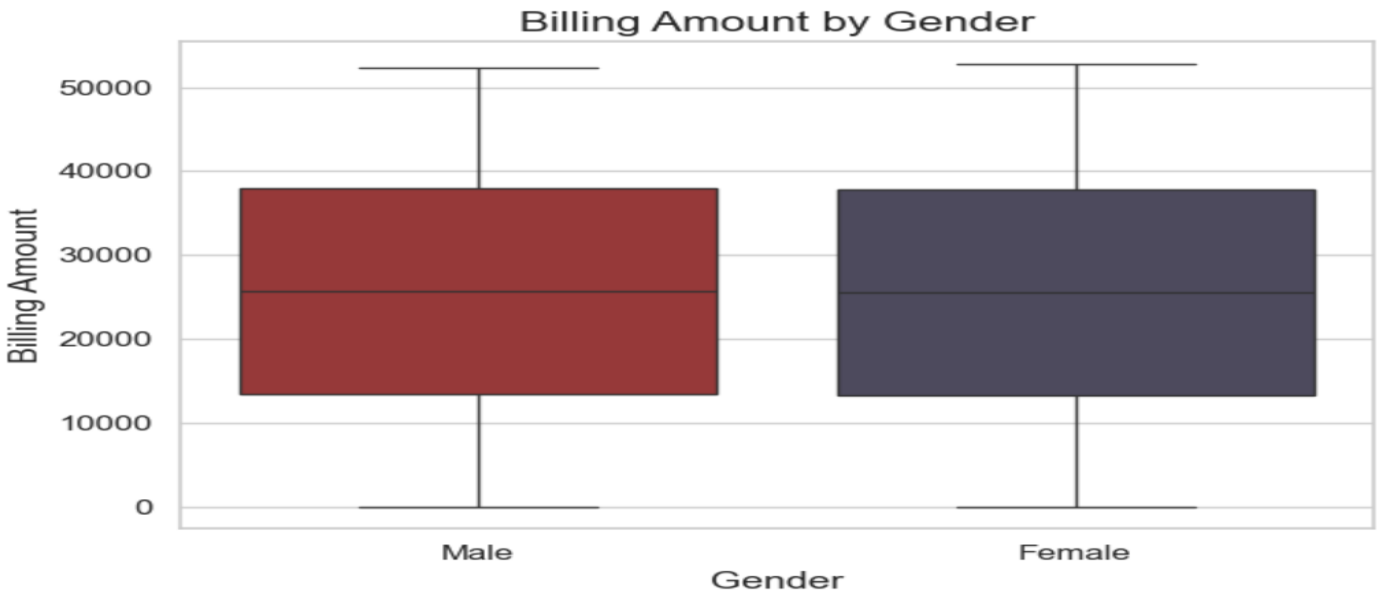


Figure 10: Billing Amount by Gender

From the boxplot, it could be seen that:

- Both sexes showed similar median billing amounts, with no significant gender discrimination. The almost exact interquartile range reflects equal healthcare expenses between sexes.

Grouped Bar Charts of Categorical Interdependencies:

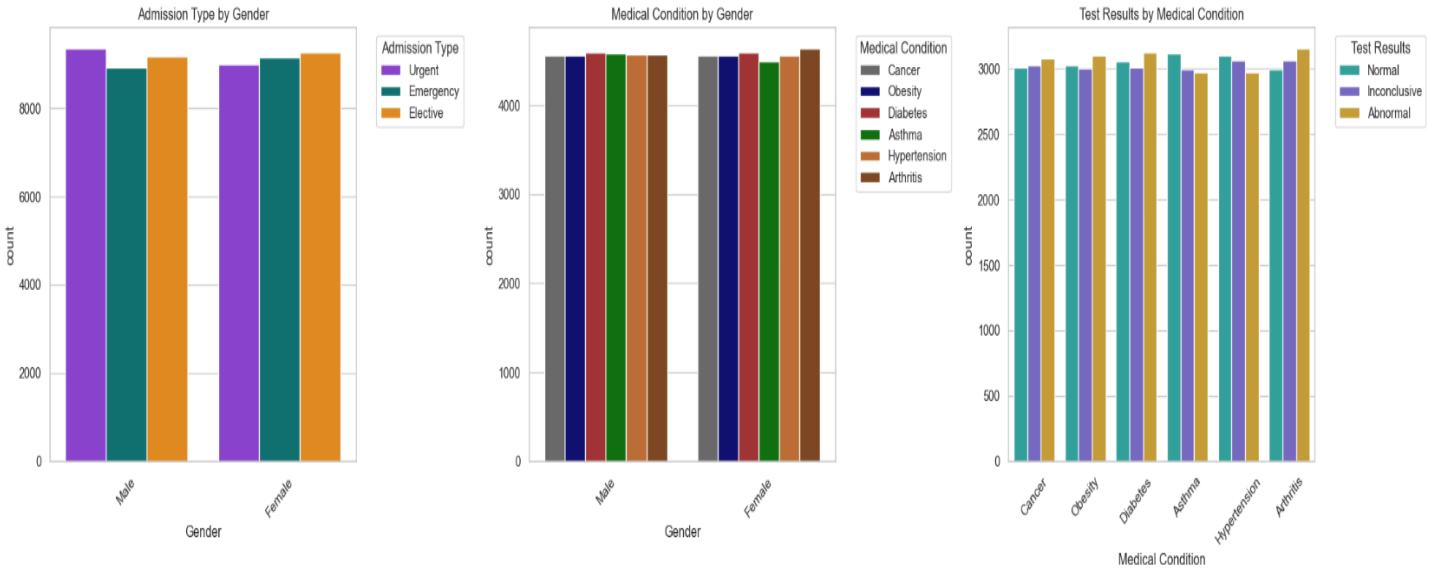


Figure 11: Categorical Interdependencies

Key Insights from the Figure 10 above are:

- **Admission Type by Gender:** Although elective and urgent admissions were more similarly distributed between sexes, showing small gender-based variations in healthcare use, males had a relatively higher proportion of emergency admissions.
- **Medical Condition by Gender:** Higher frequency of chronic diseases like obesity, hypertension, and arthritis among women indicated possible gender-connected health hazards.
- **Test Results by Medical Conditions**: Although normal results were usually seen in asthma and arthritis occasions, indicating the different degree and character of medical disorders, abnormal test outcomes were most common among cancer and diabetes patients.

Correlation Matrix of the Numerical Variable:

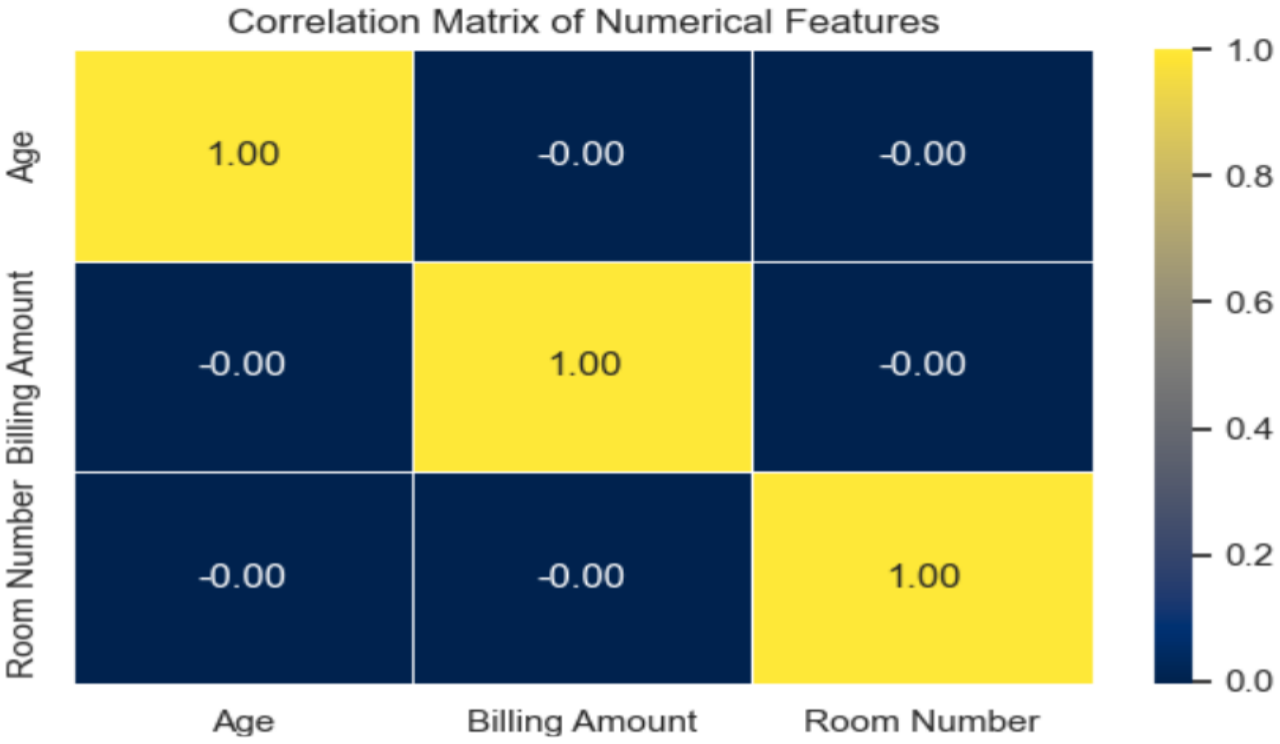


Figure 12: Correlation Heatmap of Numerical Features

Key Insights from the Figure 10 above are:

- **Age Vs Room Number:** With almost 0% association, the room given has no bearing on the age of a patient. This is expected as room numbers are usually allocated depending more on logistics than on demographic information.
- **Age Vs Billing Amount:** The billing amount and a patient's age have a somewhat minimal positive connection. This implies that age is not a good indicator of billing even although older patients might have somewhat more healthcare expenses.
- **Billing Amount Vs Room Number:** Likewise, the billing amount has no correlation with room counts, thereby verifying that room distribution is free from charging category.

Time-Based Analysis:

This is used for displaying pattern across time, increasing trends' visibility, consistency and highlighting rises and reductions during time.

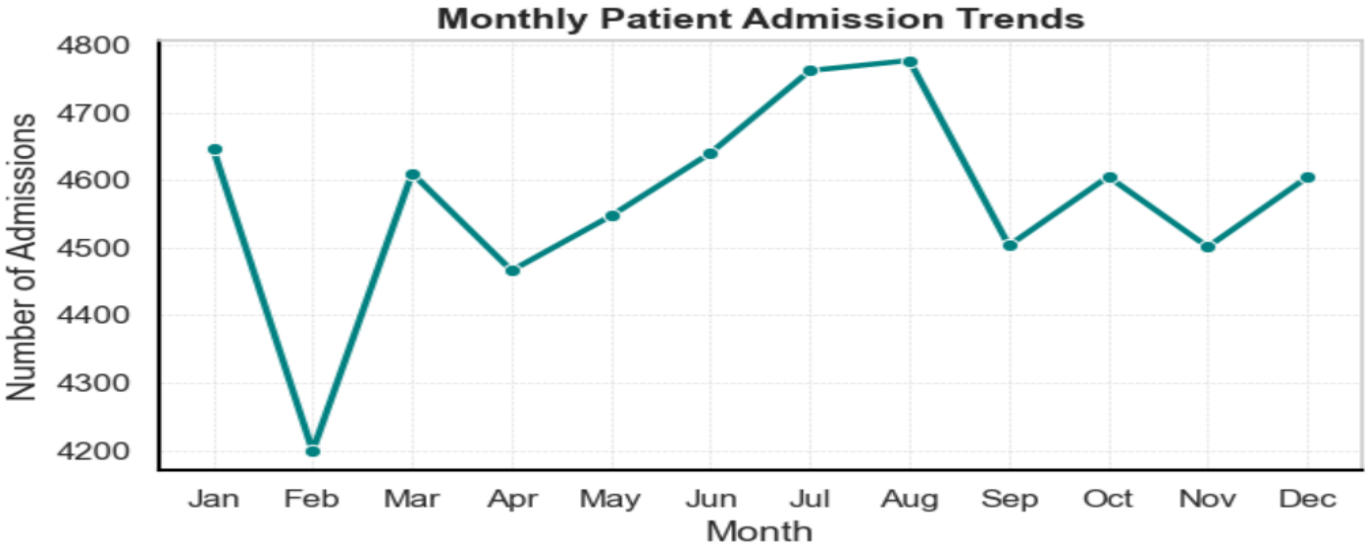


Figure 13: Patients Admissions Trend Over Different Months

Key Insights from the Figure 13 above are:

- Patient admissions clearly exhibit changes in the pattern line all year long.
- August had the most admissions—4,832 patients—while February had the lowest—4,255 patients—suggesting a possible seasonal trend.
- These trends point to chances for improved hospital resource planning such as staff assignment, service preparation, and cost-effective resource allocation matched with flow of patients.

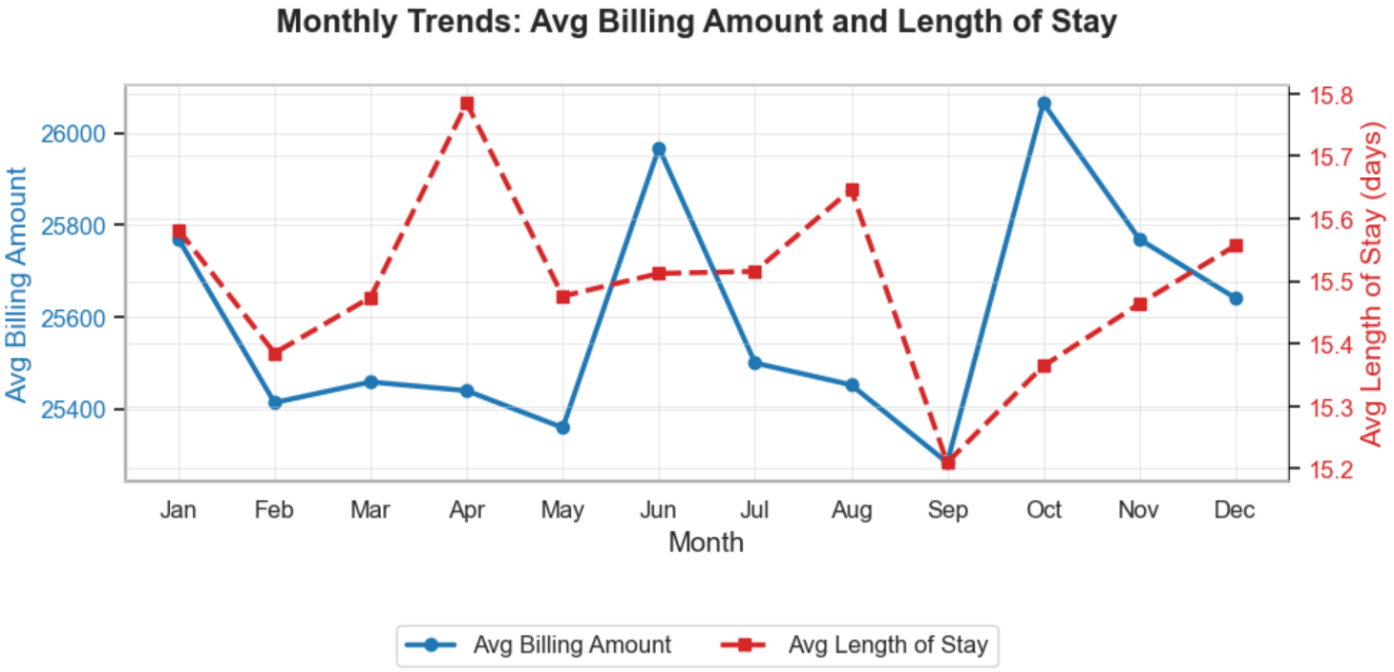


Figure 14: Average Billing Amount and the Length of Stay

Key Insights from the Figure 14 above are:

- With the highest average billing, October indicated either more extensive treatments or rising service expenses during this time.
- April's average hospital stay, on the other hand, was the longest, suggesting a possible increase in seasonal or chronic conditions needing more prolonged treatment.
- Fascinatingly, the non-aligning peaks suggest that the kind or complexity of the services rendered rather than longer hospital stays invariably determines greater billing.
- In healthcare environments, these realizations help to promote better budgetary planning, seasonal staffing policies, and allocation of resources.

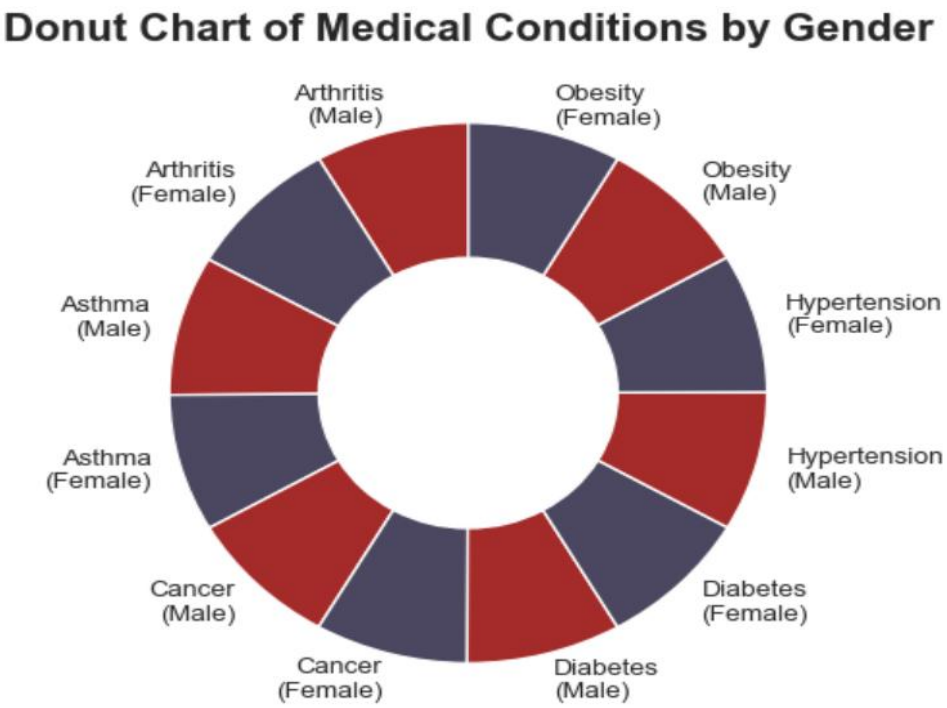


Figure 15: Donut Chat of Medical Condition by Gender

Key Insights from the Figure 15 above are:

- The graphic demonstrates a balanced gender distribution across most medical disorders, with a few significant exceptions.
- Some illnesses (e.g., Hypertension, Diabetes) have a bigger share among one gender, indicating possible gender-specific risk patterns. This division is critical for guiding focused healthcare initiatives and knowing how gender may influence illness incidence and treatment trends.

Plotly Express

Plotly express is an advanced data visualization tool enabling very minimal coding to produce interactive graphs. Built atop Plotly Graph Objects, it provides a lower-level interface for creating customized visualizations (Cotton, 2022). Some charts are created using plotly express with the use of python programming language as shown below:

```python
# Plotly express library imported as 'xp'
import plotly.express as xp
fig = xp.sunburst( # Sunburst chart created using the DataFrame 'df'
    df,  # The DataFrame containing the data
    path=['Admission Type', 'Medical Condition', 'Gender'], # The degree of categories to show in the sunburst
    values=None,
    color='Gender',
    color_discrete_map={ # Custom colors assigned to male and female variables
        'Male': 'brown',
        'Female': '#4B4760'
    },
    title='Sunburst Chart of Admission Type,Medical Condition, and  Gender Distribution', width=800, height=600)
fig.show() #Outputing the surnburst chart
```

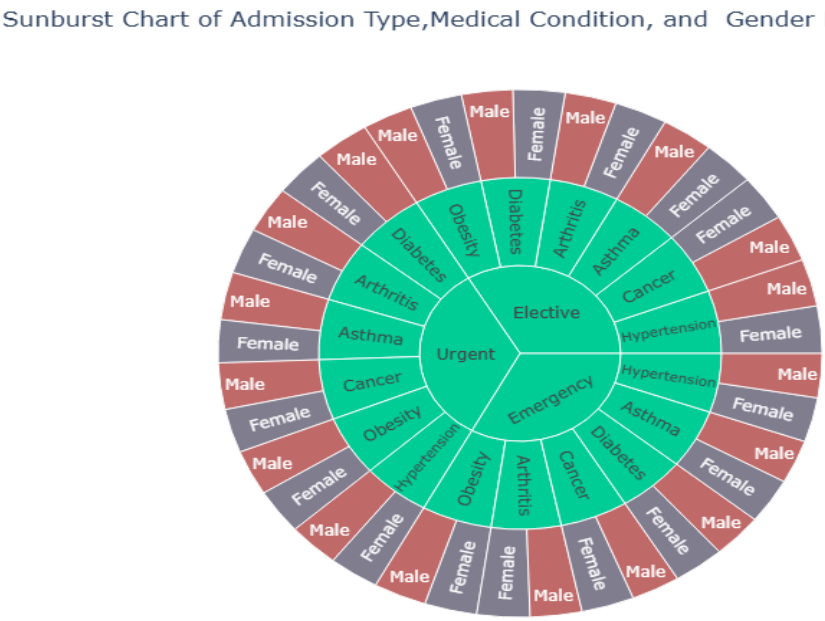


Figure 16: Sunburn Chart of Admission Type, Medical Condition, and Gender Distribution

Key Insights from the Figure 16 above are:

- ❖ Driven mostly by chronic lifestyle disorders including hypertension, diabetes, and obesity, urgent admissions emphasize a significant need for urgent medical services.
- ❖ Gender patterns differ between diseases; although cancer and asthma exhibit a more balanced distribution, women are more likely to be found in cases of arthritis and obesity; these insights can direct focused treatments and healthcare resource allocation

Animated Chart:



Figure 17: Monthly Patients Admissions Over Years

Key Insights from the Figure 15 above are:

- ❖ Although February usually has the lowest patient flow, hospital admissions show stable seasonal tendencies; peak numbers usually occur in August and October. These trends imply fundamental seasonal effects on demand for healthcare like climate, holidays, or scheduling cycles.
- ❖ The year-over-year conformity in monthly patterns gives hospitals a solid basis for strategic decision-making involving allocation of resources, personnel deployment, and focused public health campaigns during abundant supply periods.

CHAPTER FIVE
DATA VISUALIZATION AND ANALYSIS

5.1 Techniques for Visualization:

In this chapter, several visualization strategies are investigated to find patterns, correlations, and anomalies in the healthcare dataset. Initial analyses were undertaken using basic plots such as histograms and bar charts to assess univariate distributions of data. These methodologies showed crucial characteristics, such as a greater proportion of adult patients, skewed distributions in billing amounts, and a reasonably gender balance.

Bivariate analysis provided crucial insights using multiple visualizations: scatter plots demonstrated the heterogeneity in healthcare expenses over age groups, violin plots revealed billing distributions by medical condition, and box plots compared billing trends within genders. Also, a heatmap of association coefficients indicated a slight correlation between age and both billing amount and room number.

Grouped bar charts showed the connections among category variables, indicating gender-related patterns in admission types and medical problems. Advanced visuals with interaction utilizing Plotly Express—such as sunburst, animated bar, and donut charts—enabled dynamic, multi-level data analysis, making trends and insights easier to identify.

5.2 Data Storytelling and Insight Development:

The following stories and insights were discovered in course of the data visualization and analysis.

- The primary value of these visualizations rests in their capacity to tell stories with data. The **"sunburst"** graphic clearly highlighted the rank of **"Admission Type"** → **"Medical Condition"** → **"Gender"**. It indicated that urgent admissions were most prevalent and generally involved chronic illnesses such as **"obesity"**, **"hypertension"**, and **"diabetes"**. Gender distribution across these categories suggests trends that might impact healthcare development.
- The animated bar chart highlighted seasonal changes in patient enrollments, with regular peaks in **"August"** and **"October"** and minimums in **"February"**. This pattern, observed throughout numerous years, implies the presence of seasonal influences that affect healthcare demand. Hospitals can use these findings for more efficient resources and staff scheduling.
- **"Donut charts"** and other category-based visualizations highlighted gender-related trends in disease prevalence and test outcomes. For example, the graphic representation of medical condition by gender indicated a greater number of women suffering from **"obesity"** and **"arthritis"**, indicating possible gender-based health concerns.
- All visualizations contributed to a wider story about hospital operations, patient demographics, and financial implications. These stories offer healthcare administrators and policymakers with significant insights that go beyond raw data—delivering evidence-based assistance in planning, budget allocation, and intended medical interventions.

CHAPTER SIX
LIMITATIONS AND RECOMMENDATIONS

Dataset Limitations

Although the dataset gave useful insights into hospital activities and patient demographics, there are some limitations that may have affected the results:

- **Lack of Geographic Data:** The dataset failed to include regional identifiers or patients' geographical locations. This inhibited the ability to carry out spatial investigations or comprehend regional differences in healthcare delivery.
- **Likely Sampling Bias:** The dataset is collected from a public site and may not represent a properly randomly assigned sample of all hospital admissions. Certain conditions or demographic groupings can be excessively or not adequately represented.

- **Lack of Clinical Outcome:** Tracking: The dataset lacks outcome features such as readmissions, recovery rates, or mortality. As a result, the research is confined to descriptive patterns and financial data without analysing care performance.

Analytical Barriers

- The dataset reflects a static period and does not consider for real-time or long-term data. Information collected may not represent current medical developments or upcoming disorders.
- While complex visualizations were applied, certain findings might have been influenced by chart restrictions such as overplotting e.g., scatter plots or label overflow, in bar charts.

Recommendations for Future Research

The following suggestions are meant to improve the depth and relevance of next studies:

- Accurate timing admission, discharge, and service length would help one understand trends in length-of-stay and peak service hours.
- Building on the developed visualizations here, healthcare professionals might use a live or refreshable dashboard for continuous tracking and decision-making.
- Trend normalization and validation would be made possible by a comparison between the dataset and national healthcare statistics or census data.
- Adding city, state, or GPS-level location data would allow geographical health analysis, helping hospitals to organize services based on regional demand trends.
- Including outcome metrics such as treatment success, discharge status, or follow-up visit counts would assist correlate medical problems with outcomes for patients.

CHAPTER SEVEN
LEGAL AND ETHICAL CONSIDERATIONS

Management of healthcare data needs careful compliance to legal and ethical standards to assure adherence, protect patient privacy, and retain public trust. This chapter focuses on current UK legislation and essential ethical standards, particularly as defined by the British Healthcare Business Intelligence Association (BHBIA, 2023).

Legal Frameworks Governing Healthcare Dataset

Healthcare datasets sometimes include sensitive data and are regulated by the UK General Data Protection Regulation (UK GDPR) and the Data Protection Act 2018, which govern how data is obtained, handled, stored, and disseminated. To achieve compliance, healthcare analytics must be based on a valid legal rationale, such as permission, legitimate interest, or contractual requirement (BHBIA, 2023).

Data protection regulations distinguish between personal data and unique category data, such as health information, which normally requires specific agreement for usage. However, if the data is effectively

anonymized or pseudonymised, it may no longer adhere to these regulations, depending on the precise circumstances.

Data Security and Ethical Use

Ethical tackling of healthcare data extends beyond legal requirements, concentrating on respectful and responsible use. This involves being upfront about data gathering aims, reducing the quantity of data gathered, and avoiding misuse. Analysts must guarantee that data is not used in ways that might deceive or harm persons. Data security is critical, needing strong organizational and technical safeguards to avoid breaches. This includes safe storage, limited access, and encrypted data exchanges. Data processors must follow rigorous contractual duties, and organizations should be prepared to perform data protection impact assessments when handling high-risk data operations.

Individual Rights and Consent

Informed consent is frequently required when processing identifiable data, which means that individuals must be aware of the data being collected, its intended use, and its purpose. They also have rights under the UK GDPR, including the ability to access, amend, delete, and object to the use of their personal data. Organizations must maintain records of consent, manage consent in a transparent manner, facilitate withdrawal, and carry out regular audits to ensure compliance.

Ethical Data Practice Recommendations

- For high-risk data processing, perform data protection impact assessments (DPIAs)
- Keep consent management systems and privacy notices current.
- Wherever possible, anonymize or pseudonymize data.
- Keep data in jurisdictions that adhere to adequate data protection standards.
- Create contracts that specify each data handler's duties and responsibilities in detail.
- Regularly train employees who handle personal data.

To sum up, working with healthcare datasets necessitates a deep comprehension of the associated ethical and legal obligations. Compliance and the ethical stewardship of sensitive data are guaranteed when BHBIA's guidelines and the UK GDPR's legal requirements are followed.

CHAPTER EIGHT
CONCLUSION

This assessment investigated how methods of data visualization might expose important information from a large-scale healthcare database. Starting with careful data cleaning to guarantee accuracy by deleting duplicates, fixing erroneous entries, and standardizing forms, the study proceeded to combining sophisticated tools like sunburst charts and animated bar graphs with basic visualizations including histograms and bar charts, exploratory data analysis (EDA), by revealing seasonal admission patterns, gender-specific conditions, and provider billing discrepancies, these methods improved data interpretation and narrative.

Important revelations were the predominance of urgent admissions, the prevalence of chronic diseases including diabetes and hypertension, and gender-based patterns in diagnosis including obesity and arthritis. Images also underlined how a few factors greatly affect the cost of healthcare.

The dataset lacked geographic markers, time-of-day data, and patient outcome measures among other things. Suggestions centered on enhancing temporal resolution, tracking clinical results, and including location data.

Following UK GDPR and BHBIA policies guaranteed privacy, openness, and responsible data use, so addre ssing legal and ethical issues.

All things considered; this initiative shows how valuable visual analytics are in the medical field. Applied sensibly, these instruments can support evidence-based decision-making, enhance patient care, and help to improve service delivery.

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APPENDICES

Appendix A

1. Data Cleaning:

Necessary libraries importation and reading of the “health_care.csv” dataset

```
# Importing neccessary librays
import numpy as np
import scipy as sp
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore", category=FutureWarning)
```

```
df = pd.read_csv("healthcare_dataset.csv") #Loading the dataset
```

Display the first few rows of the dataset

```
df.head() # Display the first few rows
```

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
0	Bobby JacksOn	30	Male	B-	Cancer	2024-01-31	Matthew Smith	Sons and Miller	Blue Cross	18856.281306	328	Urgent	2024-02-02	Paracetamol	Normal
1	LesLie TErRy	62	Male	A+	Obesity	2019-08-20	Samantha Davies	Kim Inc	Medicare	33643.327287	265	Emergency	2019-08-26	Ibuprofen	Inconclusive
2	DaNnY sMitH	76	Female	A-	Obesity	2022-09-22	Tiffany Mitchell	Cook PLC	Aetna	27955.096079	205	Emergency	2022-10-07	Aspirin	Normal
3	andrEw waTtS	28	Female	O+	Diabetes	2020-11-18	Kevin Wells	Hernandez Rogers and Vang,	Medicare	37909.782410	450	Elective	2020-12-18	Ibuprofen	Abnormal
4	adrIENNE bEll	43	Female	AB+	Cancer	2022-09-19	Kathleen Hanna	White-White	Aetna	14238.317814	458	Urgent	2022-10-09	Penicillin	Abnormal

Displays the list of all the columns names and information about the dataset.

```
print("\nColumn Names:", df.columns.tolist())  # List of all the features names
```

```
Column Names: ['Name', 'Age', 'Gender', 'Blood Type', 'Medical Condition', 'Date of Admission', 'Doctor', 'Hospital', 'Insurance Provider', 'Billing Amou
nt', 'Room Number', 'Admission Type', 'Discharge Date', 'Medication', 'Test Results']
```

```
df.info() # Information of the dataset is displayed
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 55500 entries, 0 to 55499
Data columns (total 15 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Name                55500 non-null  object
 1   Age                 55500 non-null  int64
 2   Gender              55500 non-null  object
 3   Blood Type          55500 non-null  object
 4   Medical Condition   55500 non-null  object
 5   Date of Admission   55500 non-null  object
 6   Doctor              55500 non-null  object
 7   Hospital            55500 non-null  object
 8   Insurance Provider  55500 non-null  object
 9   Billing Amount      55500 non-null  float64
 10  Room Number         55500 non-null  int64
 11  Admission Type      55500 non-null  object
 12  Discharge Date      55500 non-null  object
 13  Medication          55500 non-null  object
 14  Test Results        55500 non-null  object
dtypes: float64(1), int64(2), object(12)
memory usage: 6.4+ MB
```

Data Cleaning, preprocessing, and feature engineering

```
# Standardizing the text features
df['Name'] = df['Name'].str.title()
df['Gender'] = df['Gender'].str.capitalize()
df['Doctor'] = df['Doctor'].str.title()
df['Hospital'] = df['Hospital'].str.title()
df['Insurance Provider'] = df['Insurance Provider'].str.title()
df['Medical Condition'] = df['Medical Condition'].str.title()
df['Admission Type'] = df['Admission Type'].str.title()
df['Medication'] = df['Medication'].str.title()
df['Test Results'] = df['Test Results'].str.capitalize()
df['Blood Type'] = df['Blood Type'].str.upper()

# Converting date features to Datetime
df['Date of Admission'] = pd.to_datetime(df['Date of Admission'], errors='coerce')
df['Discharge Date'] = pd.to_datetime(df['Discharge Date'], errors='coerce')

#Dropping duplicates data points
current_shape = df.shape
df.drop_duplicates(inplace=True)
shape_after_dropping = df.shape
```

```python
# Checking for outliers with IQR method
numeric_cols = ['Age', 'Billing Amount', 'Room Number']
outlier_summary = {}

for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]
    outlier_summary[col] = {
        'lower_bound': lower_bound,
        'upper_bound': upper_bound,
        'num_outliers': outliers.shape[0]
    }

# Showing the shapes before and after dropping duplivcates, and outlier summary
current_shape, shape_after_dropping, outlier_summary
```

Summary table for categorical variables

```python
# Creating a summary for categorical features
summary_data = []
for col in df.select_dtypes(include='object').columns:
    unique_vals = df[col].unique()
    sample_vals = ', '.join(map(str, unique_vals[:5])) + ('...' if len(unique_vals) > 5 else '')
    summary_data.append({
        'Column': col,
        'Unique Values': len(unique_vals),
        'Sample Values': sample_vals
    })
#Display the DataFrame Summary
summary_df = pd.DataFrame(summary_data)
summary_df
```

	Column	Unique Values	Sample Values
0	Name	49992	Bobby JacksOn, LesLie TErRy, DaNnY sMitH, andr...
1	Gender	2	Male, Female
2	Blood Type	8	B-, A+, A-, O+, AB+...
3	Medical Condition	6	Cancer, Obesity, Diabetes, Asthma, Hypertensio...
4	Date of Admission	1827	2024-01-31, 2019-08-20, 2022-09-22, 2020-11-18...
5	Doctor	40341	Matthew Smith, Samantha Davies, Tiffany Mitche...
6	Hospital	39876	Sons and Miller, Kim Inc, Cook PLC, Hernandez ...
7	Insurance Provider	5	Blue Cross, Medicare, Aetna, UnitedHealthcare,...
8	Admission Type	3	Urgent, Emergency, Elective
9	Discharge Date	1856	2024-02-02, 2019-08-26, 2022-10-07, 2020-12-18...
10	Medication	5	Paracetamol, Ibuprofen, Aspirin, Penicillin, L...
11	Test Results	3	Normal, Inconclusive, Abnormal

2. Exploratory Data Analysis and Visualizations:

```python
df.describe().T # Displaying the summary statistics of the cleaned dataset
```

	count	mean	min	25%	50%	75%	max	std
Age	54966.0	51.535185	13.0	35.0	52.0	68.0	89.0	19.605661
Date of Admission	54966	2021-11-01 17:35:29.505512448	2019-05-08 00:00:00	2020-07-28 00:00:00	2021-11-02 00:00:00	2023-02-03 00:00:00	2024-05-07 00:00:00	NaN
Billing Amount	54966.0	25544.30626	-2008.49	13243.7225	25542.75	37819.8575	52764.28	14208.409715
Room Number	54966.0	301.124404	101.0	202.0	302.0	401.0	500.0	115.223143
Discharge Date	54966	2021-11-17 05:34:28.202161408	2019-05-09 00:00:00	2020-08-13 00:00:00	2021-11-18 00:00:00	2023-02-19 00:00:00	2024-06-06 00:00:00	NaN

```python
df = df[df['Billing Amount'] >= 0] # Removing all the billing amount containing negative values
```

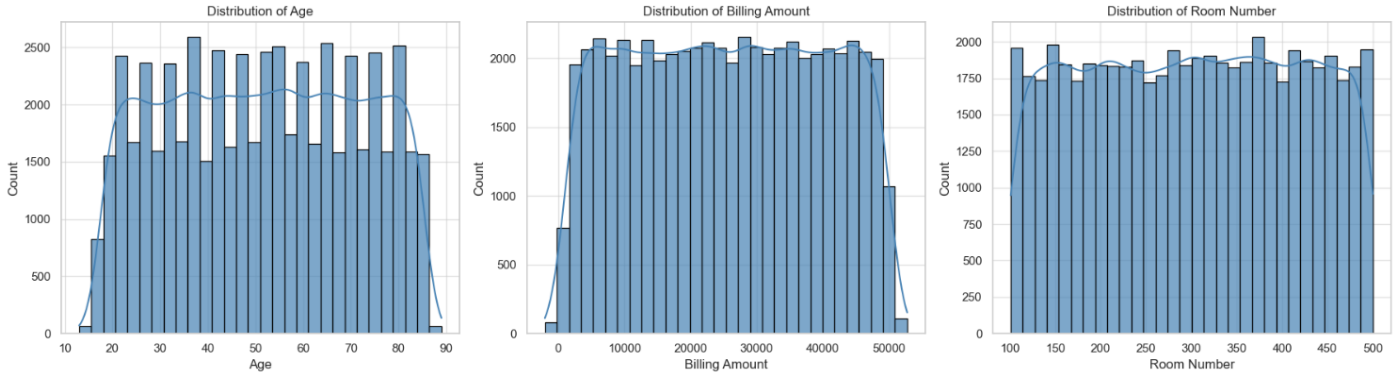
```python
df.describe().T
```

	count	mean	min	25%	50%	75%	max	std
Age	54860.0	51.53385	13.0	35.0	52.0	68.0	89.0	19.605295
Date of Admission	54860	2021-11-01 19:01:50.506744576	2019-05-08 00:00:00	2020-07-28 00:00:00	2021-11-02 00:00:00	2023-02-03 00:00:00	2024-05-07 00:00:00	NaN
Billing Amount	54860.0	25594.633612	9.24	13299.75	25593.875	37847.065	52764.28	14175.867046
Room Number	54860.0	301.109752	101.0	202.0	302.0	400.0	500.0	115.217195
Discharge Date	54860	2021-11-17 07:00:08.137076224	2019-05-09 00:00:00	2020-08-13 00:00:00	2021-11-18 00:00:00	2023-02-19 00:00:00	2024-06-06 00:00:00	NaN

Subplots distribution of numerical variables created

```
# Set seaborn style
sns.set(style="whitegrid")
# Define numerical columns
numerical_columns = ['Age', 'Billing Amount', 'Room Number'] # Define numerical columns
# Setting up the plot grid
fig, axes = plt.subplots(1, 3, figsize=(18, 5)) # 1 row, 3 columns
# Creat#ing histograms with KDE
for i, col in enumerate(numerical_columns): # Creating histograms with KDE
    sns.histplot(
        data=df,
        x=col,
        kde=True,
        color="steelblue",
        ax=axes[i],
        bins=30,
        edgecolor='black',
        alpha=0.7
    )
    axes[i].set_title(f"Distribution of {col}")  #Give unique title per plot
    axes[i].set_xlabel(col)
    axes[i].set_ylabel("Count")
plt.tight_layout()
plt.show()
```

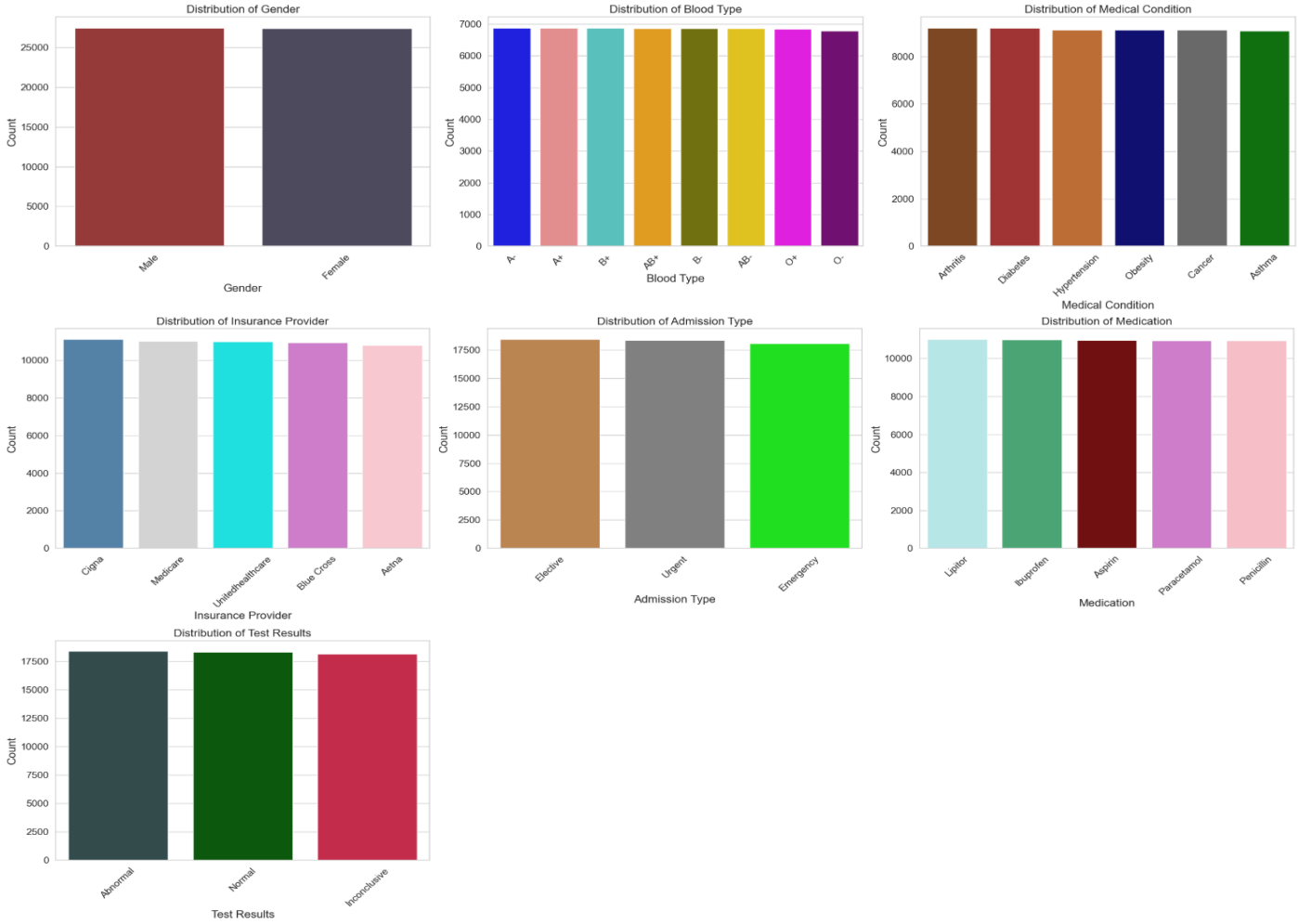
Code output



Categorical features subplots plotted in 3x3 grid

```
# Seaborn style setting
sns.set(style="whitegrid")
# Define custom color palettes for each categorical variable
custom_palettes = {
    'Gender': ['brown', '#4B4760'],
    'Blood Type': ['blue', 'lightcoral', 'mediumturquoise', 'orange', 'olive', 'gold', 'magenta', 'purple'],
    'Medical Condition' : ['saddlebrown', 'firebrick', 'chocolate', 'navy', 'dimgray', 'green'],
    'Insurance Provider': ['steelblue', 'lightgray', 'aqua', 'orchid', 'pink', 'maroon'],
    'Admission Type': ['peru', 'gray', 'lime'],
    'Medication': ['paleturquoise', 'mediumseagreen', 'maroon', 'orchid', 'lightpink'],
    'Test Results': ['darkslategray', 'darkgreen', 'crimson']
}
# Define the categorical columns for univariate analysis
categorical_columns = list(custom_palettes.keys())
# Setting up subplots
fig, axes = plt.subplots(nrows=3, ncols=3, figsize=(25, 20))
axes = axes.flatten()
# Plotting each categorical feature with the specified custom palette
for i, column in enumerate(categorical_columns):
    unique_values = df[column].value_counts().index
    palette = custom_palettes[column][:len(unique_values)]
    sns.countplot(data=df, x=column, ax=axes[i], palette=palette, order=unique_values)
    axes[i].set_title(f'Distribution of {column}', fontsize=16)
    axes[i].tick_params(axis='x', rotation=45, labelsize=14)  # Increased x-axis label size
    axes[i].tick_params(axis='y', labelsize=14)  # Increased y-axis label size
    axes[i].set_xlabel(column, fontsize=16)  # Increased x-axis title size
    axes[i].set_ylabel('Count', fontsize=16)  # Increased y-axis title size
# Hiding unused subplot
for m in range(len(categorical_columns), len(axes)):
    fig.delaxes(axes[m])
plt.tight_layout()
plt.show()
```

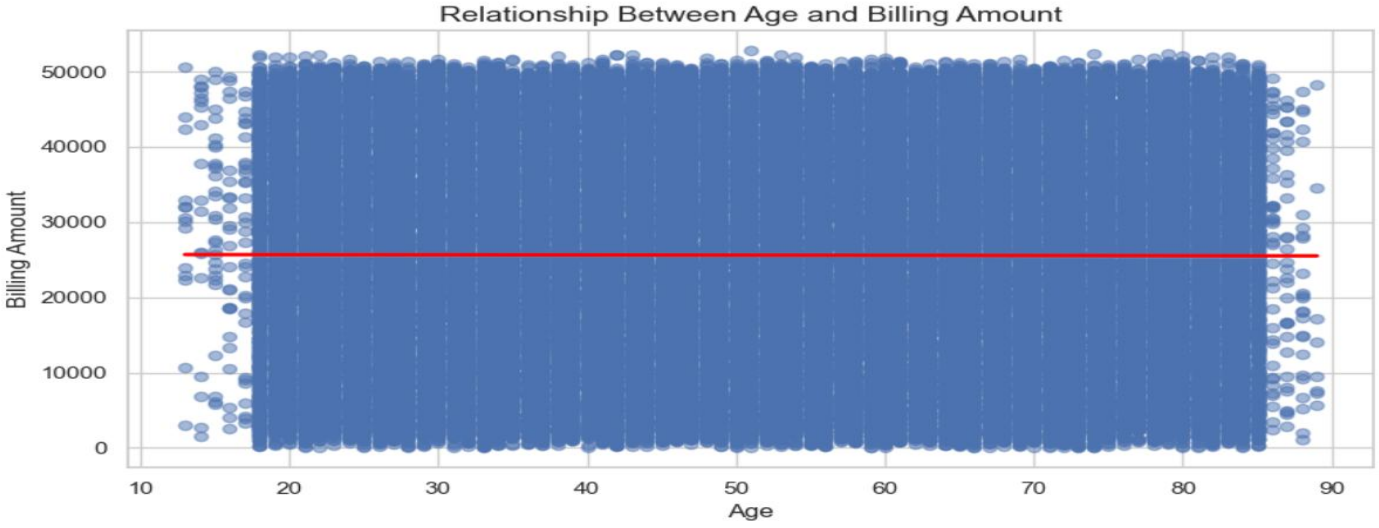
Code Output



Scatter plot that shows the relationship between Age and Billing amount

```python
# Scatter plot that shows the relationship between Age and Billing amount
sns.set(style="whitegrid")
plt.figure(figsize=(9, 5))
sns.regplot(data=df, x='Age', y='Billing Amount', scatter_kws={'alpha':0.5}, line_kws={'color': 'red'})
plt.title('Relationship Between Age and Billing Amount', fontsize=14)
plt.xlabel('Age')
plt.ylabel('Billing Amount')
plt.tight_layout()
# Saving the figure before displaying
plt.savefig('age_vs_billing.png', dpi=300, bbox_inches='tight')  # <--- This line saves the plot!
plt.show() # Display the figure
```

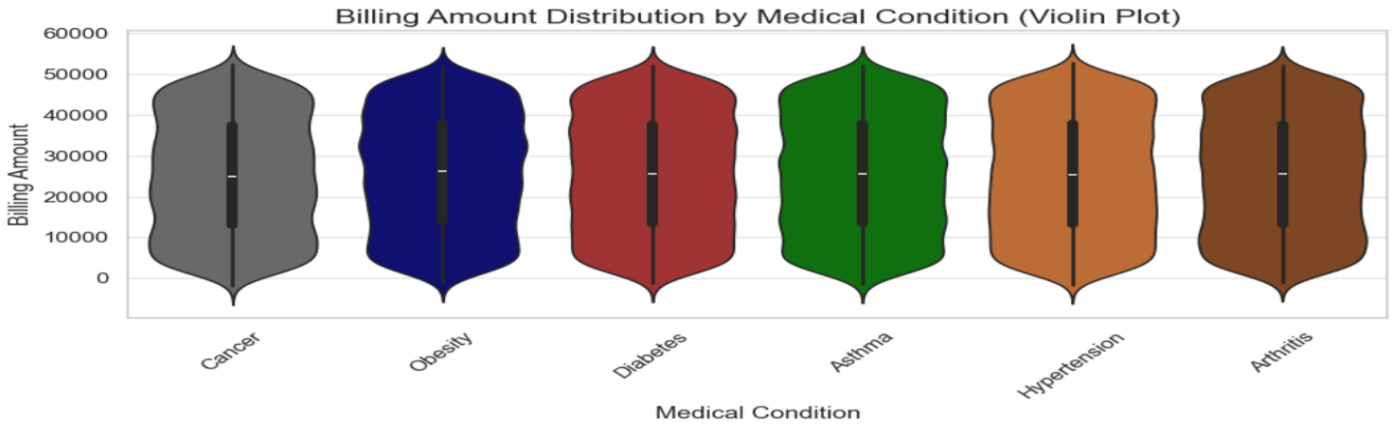
Code Output



Violin Plot

```
# Defining the various medical condition colour pallete
medical_condition_palette = {  'Cancer': 'dimgray', 'Obesity': 'navy', 'Diabetes': 'firebrick','Asthma': 'green','Hypertension': 'chocolate',
                             'Arthritis': 'saddlebrown'
}
#Plotting the violin
plt.figure(figsize=(10, 5))
sns.violinplot(data=df, x='Medical Condition', y='Billing Amount',
               palette=medical_condition_palette, order=medical_condition_palette.keys())
plt.title('Billing Amount Distribution by Medical Condition (Violin Plot)', fontsize=16)
plt.xlabel('Medical Condition', fontsize=13)
plt.ylabel('Billing Amount', fontsize=13)
plt.xticks(rotation=45, fontsize=12)
plt.yticks(fontsize=12)
plt.tight_layout()
# Saving the figure before displaying
plt.savefig('billing_by_medical_condition.png', dpi=300, bbox_inches='tight')
plt.show()
```

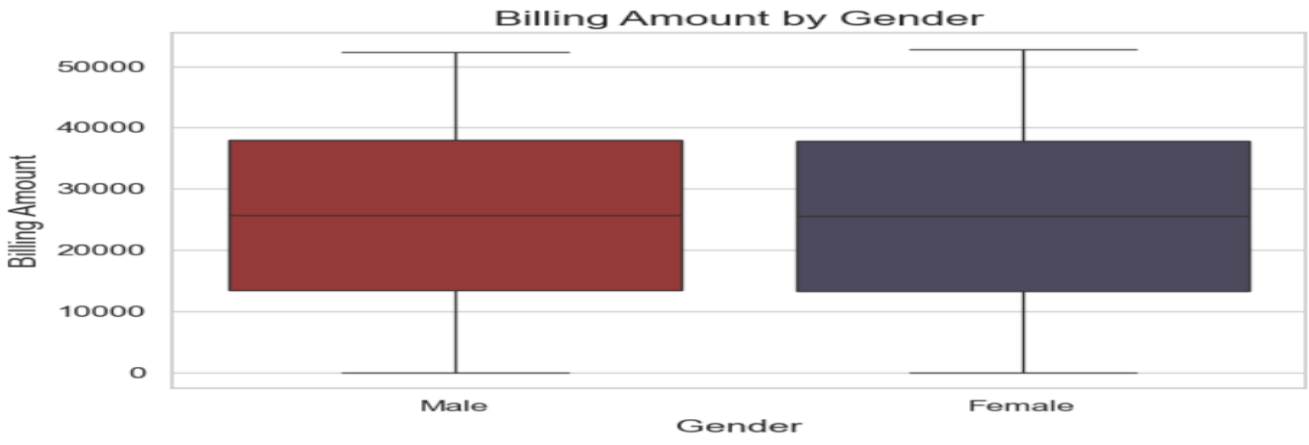
Code Output



Boxplot for Billing Among by Gender

```
# Custom colors defined for Gender
gender_palette = {'Male': 'brown', 'Female': '#4B4760'}
# Creating boxplot for Billing Amount by Gender
plt.figure(figsize=(7, 5))
sns.boxplot(data=df, x='Gender', y='Billing Amount', palette=gender_palette)
plt.title('Billing Amount by Gender', fontsize=16)
plt.xlabel('Gender', fontsize=14)
plt.ylabel('Billing Amount', fontsize=14)
plt.xticks(fontsize=12)
plt.yticks(fontsize=12)
plt.tight_layout()
# Saving the figure before displaying
plt.savefig('billing_by_gender.png', dpi=300, bbox_inches='tight')
plt.show()
```

Code Output

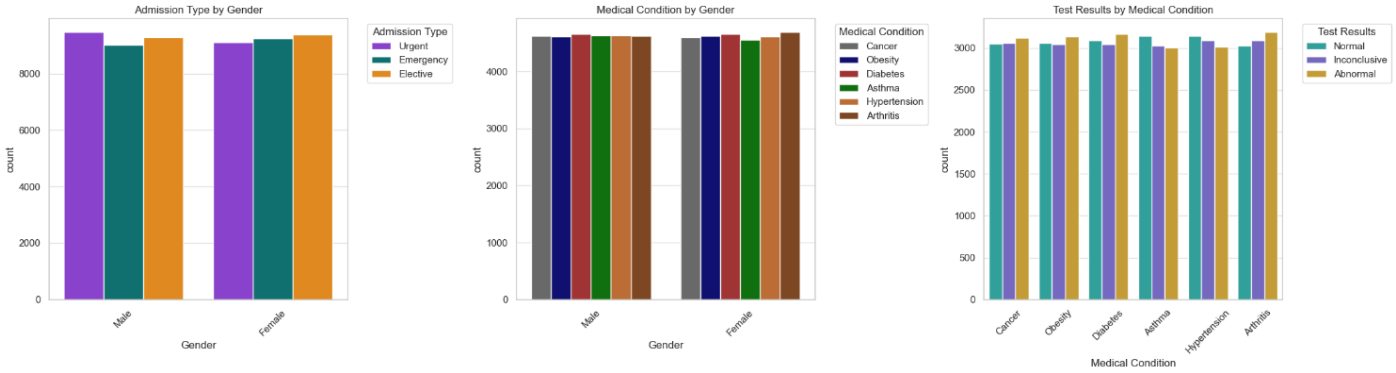


Grouped bar charts showing the relationships between pairs of categorical variables in the dataset.

```
# Grouped bar charts,custom pallete definition
admission_type_palette = {'Emergency': 'Teal',  'Elective': 'Darkorange',  'Urgent': 'Blueviolet'}
test_results_palette = { 'Normal': 'Lightseagreen',   'Abnormal': 'Goldenrod',  'Inconclusive': 'Slateblue'}
# Define the categorical pairs
categorical_pairs = [('Gender', 'Admission Type'), ('Gender', 'Medical Condition'),('Medical Condition', 'Test Results')]
# Custom condition palette defined
custom_condition_palette = { 'Cancer': 'dimgray','Obesity': 'navy', 'Diabetes': 'firebrick','Asthma': 'green', 'Hypertension': 'chocolate',
    'Arthritis': 'saddlebrown'}
# Grouped bar charts plot
fig, axes = plt.subplots(nrows=1, ncols=3, figsize=(22, 6))
axes = axes.flatten()

for i, (x_var, hue_var) in enumerate(categorical_pairs):
    if hue_var == 'Medical Condition':       # Assigning custom palettes based on the hue variable
        palette = custom_condition_palette
    elif hue_var == 'Admission Type':
        palette = admission_type_palette
    elif hue_var == 'Test Results':
        palette = test_results_palette
    else:
        palette = 'Set2'
    sns.countplot(data=df, x=x_var, hue=hue_var, ax=axes[i], palette=palette)
    axes[i].set_title(f'{hue_var} by {x_var}', fontsize=12)
    axes[i].tick_params(axis='x', rotation=45)
    axes[i].legend(title=hue_var, bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
# Save the figure before showing
plt.savefig('grouped_bar_charts.png', dpi=300, bbox_inches='tight')
plt.show()
```

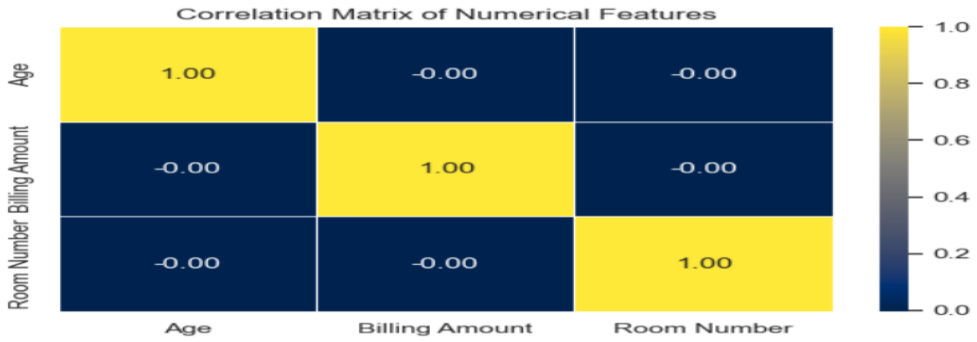
Code Output



Numerical variables correlation matrix

```
# The correlation matrix of the numerical variables
numerical_features = ['Age', 'Billing Amount', 'Room Number']
corr_matrix = df[numerical_features].corr()
plt.figure(figsize=(6, 4))  # Ploting the correlation heatmap
sns.heatmap(
    corr_matrix,
    annot=True,
    fmt=".2f",
    cmap='cividis',
    linewidths=0.5,
    linecolor='white'
)
plt.title('Correlation Matrix of Numerical Features')
plt.tight_layout()
# Saving the figure before displaying
plt.savefig('correlation_matrix.png', dpi=300, bbox_inches='tight')
plt.show()
```

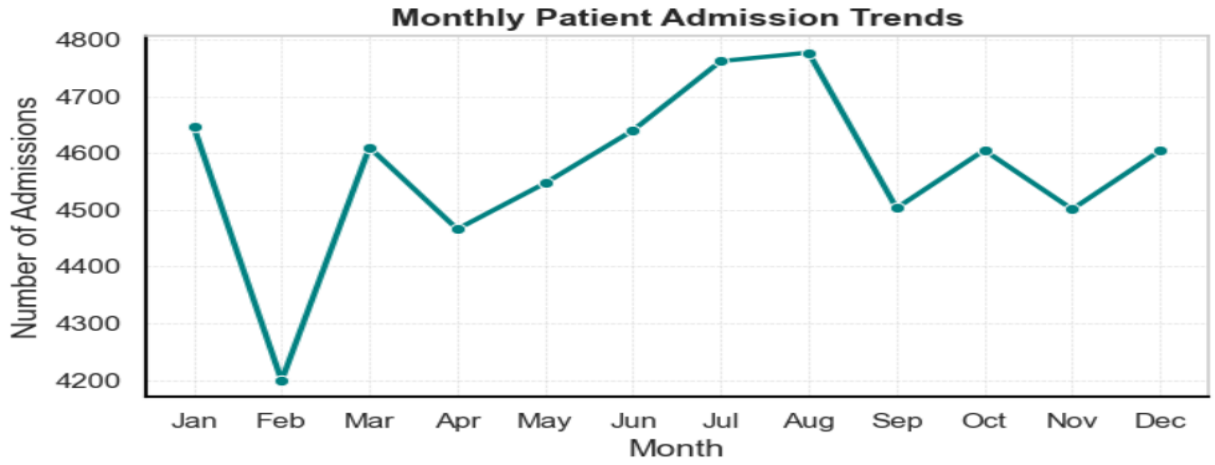
Code Output



```python
df.drop_duplicates(inplace=True) # Dropping duplicates and invalid billing datapoints
df = df[df['Billing Amount'] >= 0]
df['Length of Stay'] = (df['Discharge Date'] - df['Date of Admission']).dt.days # Computing the Length of Stay
df = df[df['Length of Stay'] > 0]  # Removing invalid stays
df['Admission Month'] = df['Date of Admission'].dt.month # Extracting time variables
df['Admission Year'] = df['Date of Admission'].dt.year
monthly_admissions = df['Admission Month'].value_counts().sort_index() # Monthly admissions count

# Seting a clean style
sns.set_style("whitegrid")
plt.rcParams.update({'axes.titlesize': 16, 'axes.labelsize': 14,'xtick.labelsize': 12,'ytick.labelsize': 12,'grid.alpha': 0.6,'grid.linestyle': '--',
    'grid.linewidth': 0.5 })
# Using enhanced axis visibility for the plot
plt.figure(figsize=(7, 4))
sns.lineplot(x=monthly_admissions.index, y=monthly_admissions.values, marker='o', linewidth=2.5, color='teal')
# Adding clear axis lines
ax = plt.gca()
ax.spines['left'].set_linewidth(1.5)
ax.spines['bottom'].set_linewidth(1.5)
ax.spines['left'].set_color('black')
ax.spines['bottom'].set_color('black')
# Adding the Titles and Labels
plt.title('Monthly Patient Admission Trends', fontsize=14, fontweight='bold')
plt.xlabel('Month')
plt.ylabel('Number of Admissions')
plt.xticks(ticks=range(1, 13), labels=['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
                                       'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'])
plt.grid(True)
plt.tight_layout()
# Save the figure before showing
plt.savefig('monthly_admissions_trend.png', dpi=300, bbox_inches='tight')
plt.show()
```

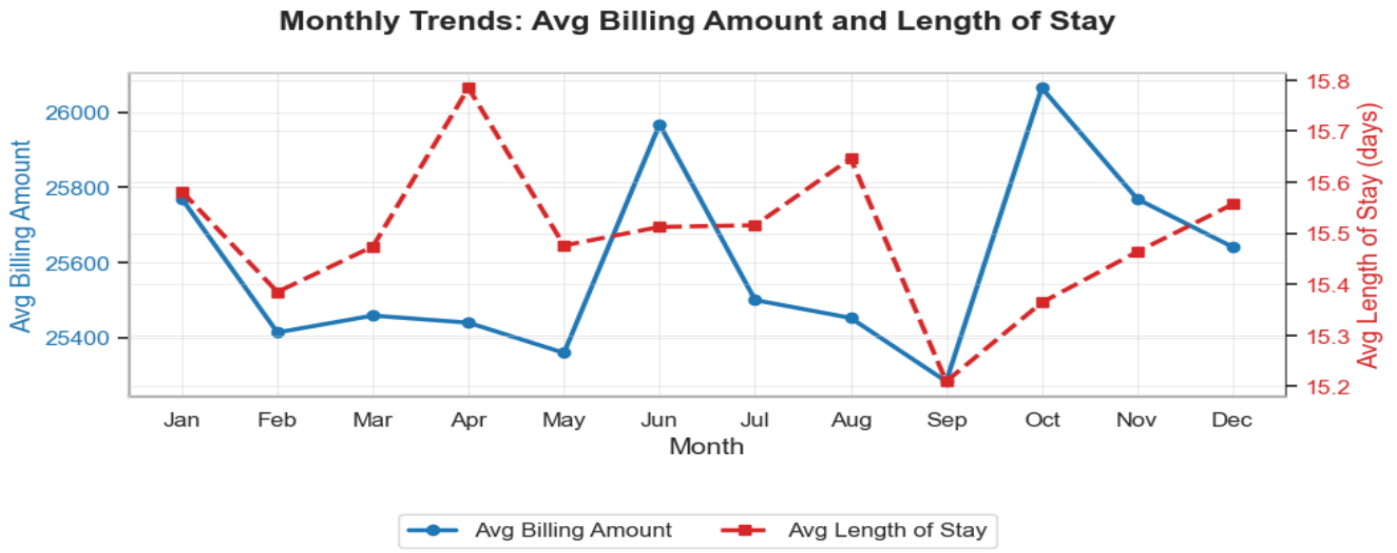
Code Output



Monthly trends of Billing Amount and Length of Stay plotted with dual axes and custom styling

```python
monthly_stats = df.groupby('Admission Month')[['Billing Amount', 'Length of Stay']].mean().reset_index()
sns.set_style("whitegrid")
fig, ax1 = plt.subplots(figsize=(10, 4))
# Styling the left and bottom axes
for spine in ['left', 'bottom']:
    ax1.spines[spine].set_linewidth(1.5)
    ax1.spines[spine].set_color('black')
# Ploting the Avg Billing Amount
color1 = 'tab:blue'
line1, = ax1.plot(monthly_stats['Admission Month'], monthly_stats['Billing Amount'],
                  color=color1, marker='o', linewidth=2.5, label='Avg Billing Amount')
ax1.set_xlabel('Month', fontsize=14)
ax1.set_ylabel('Avg Billing Amount', color=color1, fontsize=14)
ax1.tick_params(axis='y', labelcolor=color1)
ax1.set_xticks(range(1, 13))
ax1.set_xticklabels(['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun','Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'], fontsize=12)
# Plotting the Avg Length of Stay with regular right spine
ax2 = ax1.twinx()
color2 = 'tab:red'
line2, = ax2.plot(monthly_stats['Admission Month'], monthly_stats['Length of Stay'],
                  color=color2, marker='s', linestyle='--', linewidth=2.5, label='Avg Length of Stay')
ax2.set_ylabel('Avg Length of Stay (days)', color=color2, fontsize=14)
ax2.tick_params(axis='y', labelcolor=color2)
ax2.spines['right'].set_linewidth(0.5) # Restoring the right spine to default
ax2.spines['right'].set_color('black')
# Title and legend
fig.suptitle('Monthly Trends: Avg Billing Amount and Length of Stay', fontsize=16, fontweight='bold')
fig.tight_layout()
plt.grid(axis='x', linestyle='--', alpha=0.6)
fig.legend(handles=[line1, line2], loc='upper center', bbox_to_anchor=(0.5, -0.05), ncol=2, fontsize=12)
# Save the figure before showing
plt.savefig('monthly_billing_length_stay.png', dpi=300, bbox_inches='tight')
plt.show()
```

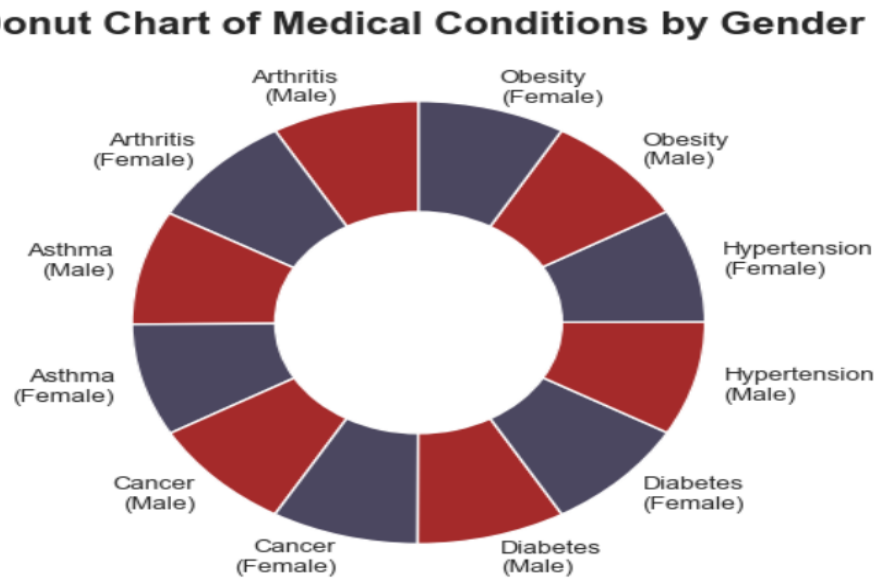
Code Output



Donut chart

```
# Count of gender within each medical condition
condition_gender_counts = df.groupby(['Medical Condition', 'Gender']).size().unstack().fillna(0)
fig, ax = plt.subplots(figsize=(5, 5)) # Setting up the plot
# Flattening the data for plotting
labels = []
sizes = []
colors = []
color_map = {'Male': 'brown', 'Female': '#4B4760'}
for condition in condition_gender_counts.index:
    for gender in ['Male', 'Female']:
        count = condition_gender_counts.loc[condition, gender]
        if count > 0:
            labels.append(f"{condition}\n({gender})")
            sizes.append(count)
            colors.append(color_map[gender])
# The donut chart created
wedges, texts = ax.pie(
    sizes, labels=labels, startangle=90, colors=colors,
    wedgeprops=dict(width=0.5, edgecolor='w'), textprops={'fontsize': 10}
)
# Adding the title
plt.title('Donut Chart of Medical Conditions by Gender', fontsize=16, fontweight='bold')
plt.tight_layout()
plt.savefig('donut_medical_conditions_gender.png', dpi=300, bbox_inches='tight')
plt.show()
```

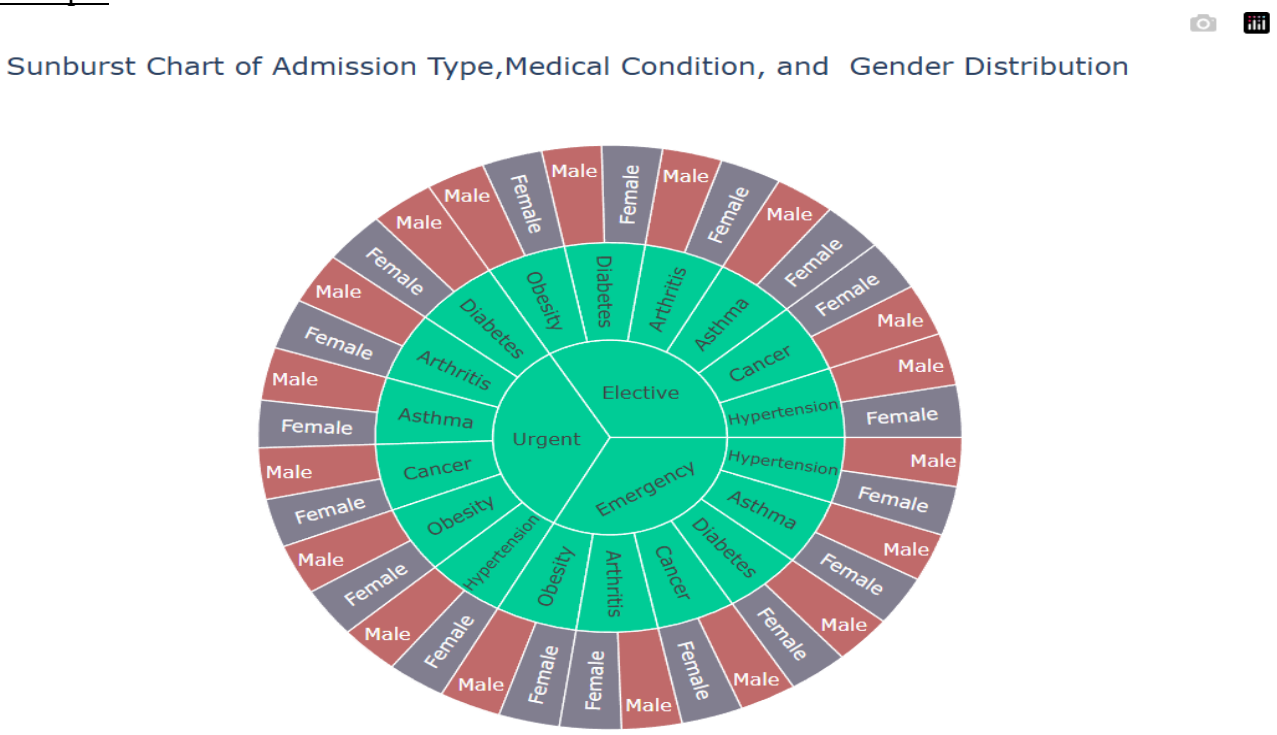
Code Output



Sunburst chart

```
# Plotly express library imported as 'xp'
import plotly.express as xp
fig = xp.sunburst( # Sunburst chart created using the DataFrame 'df'
    df,  # The DataFrame containing the data
    path=['Admission Type', 'Medical Condition', 'Gender'], # The degree of categories to show in the sunburst
    values=None,
    color='Gender',
    color_discrete_map={ # Custom colors assigned to male and female variables
        'Male': 'brown',
        'Female': '#4B4760'
    },
    title='Sunburst Chart of Admission Type,Medical Condition, and  Gender Distribution', width=800, height=600)
fig.show() #Outputing the surnburst chart
```

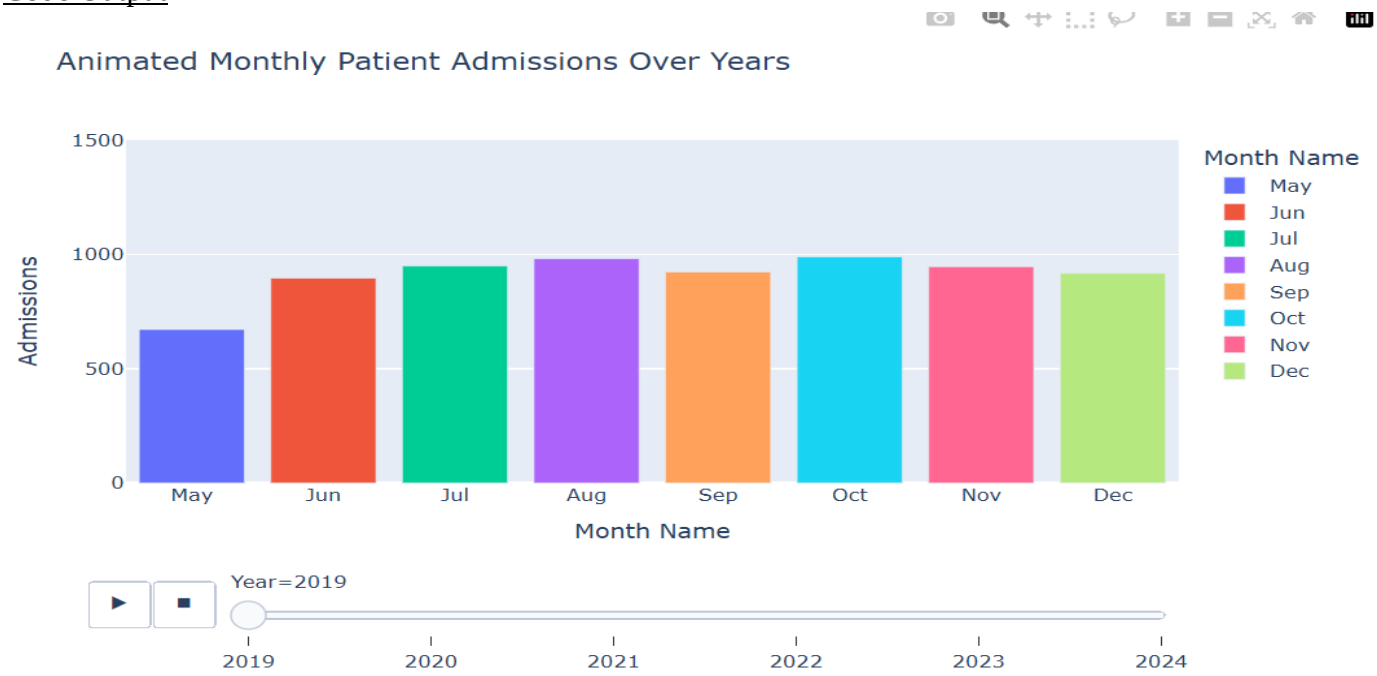
Code Output



Animated monthly admission over years

```
import calendar
# Grouping data for monthly admissions
monthly_admissions_animation = df.groupby(['Admission Year', 'Admission Month']).size().reset_index(name='Admissions')
# Converting the month number to readable labels
monthly_admissions_animation['Month Name'] = monthly_admissions_animation['Admission Month'].apply(lambda x: calendar.month_abbr[x])
monthly_admissions_animation['Year'] = monthly_admissions_animation['Admission Year'].astype(str)
# Animated bar chart generated
fig_anim = xp.bar(
    monthly_admissions_animation,
    x='Month Name',
    y='Admissions',
    color='Month Name',
    animation_frame='Year',
    title='Animated Monthly Patient Admissions Over Years',
    range_y=[0, monthly_admissions_animation['Admissions'].max() + 500],
    width=800,
    height=500
)
fig_anim.show()
```

Code Output



Appendix B

Code Execution Instructions

Python was used in a Jupyter Notebook environment for development of this project. Together with the healthcare_dataset.csv file, the entire code is available in the **240174581_DV_Code_Documentation.ipynb** file the following describes how to replicate the visualizations and study.

Requirements:

- Python 3.11.8 installed
- Jupyter Notebook
- **Required Python libraries:** Pandas, matplotlib, seaborn and plotly.

Code Execution Steps:

- Verify that the files Healthcare_Dataset.csv and **240174581_DV_Code_Documentation.ipynb** share the same directory.
- On your platform, open Jupyter Notebook.
- Lunch the **240174581_DV_Code_Documentation.ipynb** notebook.
- Execute every code cell one after the other:
  - Cleaning and loading data.
  - Features Engineering
  - Visualizations Blocks
- Visual outputs will either be saved as.png files in the working directory or shown inline.